

PROVISION OF FIRST AID SERVICES

1 4.1 Introduction to the unit of learning

This unit specifies the competencies required to provide first aid. It involves providing cardio-pulmonary resuscitation; managing choking, burns, scalds, trauma, nose bleeding, cuts, fractures, drowning, poisoning and snake bites. It also involves assisting patients with existing conditions such as asthma, epilepsy and diabetes.

2 4.2 Summary of Learning Outcomes

1. Assess the nature and extent of injury or illness
2. Provide first aid intervention
3. Evaluate first aid intervention
4. Wind-up first aid intervention

4.2.1 Learning Outcome 1:

Assess the nature and extent of injury or illness

4.2.1.1 Introduction to the learning outcome

This learning outcome specifies the content of competencies required to assess the nature and extent of injury or illness. It includes principles of first aid, scene size up, assessment of nature and extent of injury and /or illness, sourcing for emergency services, decision making-triaging and requirements for first aid. Effective assessment of the nature and extent of injury or illness reduces the mortality rate by 75% in most of emergency cases.

4.2.1.2 Performance Standard

1. Assessed personal, casualty and environment safety as per principles of first aid.
2. Assessed the nature and extent of injury and /or illness in accordance to first aid principles.
3. Called assistance from emergency services as per first aid principles.
4. Made a decision promptly regarding action to be taken.
5. Identified tools, equipment, materials and supplies required based on the assessment carried out.
6. Gathered tools, equipment, materials and supplies based on the assessment of the situation.

4.2.1.3 Information Sheet

Definition of Terms

Casualty- the department of a hospital providing immediate treatment for emergency cases

First Aid- is the first and immediate assistance given to any person suffering from either a minor or serious illness or injury, with care provided to preserve life, prevent the condition from worsening, or to promote recovery

Emergency services- are the public organizations whose job is to take quick action to deal with emergencies when they occur, especially the fire brigade, the police, and the ambulance service.

Injury- also known as physical trauma is damage to the body caused by external force. This may be caused by accidents, falls, hits, weapons, and other causes.

Principles of first aid

Preserve Life

The first aim of first aid is to preserve life, which involves the key emergency practices to ensure that the casualty isn't in any mortal danger. Remember though, this includes preserving your own life as you shouldn't put yourself in danger in order to apply first aid. It's at this stage where you should do a quick risk assessment to check for dangers to the injured person, yourself or bystanders which could cause the situation to escalate. If in doubt, do not attempt to apply first aid and immediately call for a medical professional.

Prevent Deterioration

Once you've followed all the steps associated with the first principle, your next priority is to prevent deterioration of the injured person's condition. Keeping a casualty still to avoid aggravating their injury, or from complicating any unseen issues, is crucial. This helps prevent to further injuries, and clearing the area of any immediate dangers will help you to do so.

Promote Recovery

Finally, there are steps you should follow which will help lessen the amount of time taken for a casualty to recover from an accident and aid in minimizing lasting damage and scarring. The prime example of this is applying cold water to a burn as soon as possible to lower the chance of long-term scarring and helps speed up the healing process.

Scene Size-up

The scene size-up is performed as units first approach their call. While the activity is part of most EMS training programs, it is often limited to the student asking: "I perform my scene size-up, what do I see?" and then being given answers by a well-meaning instructor looking to move the scenario along to the "important clinical part."

On the job, very few calls present consequences for the responder if a thorough scene size-up is not performed; as a result, providers become complacent and out of practice. All skills in medicine fade with disuse and many providers weren't proficient at a scene size-up to begin with. More often than not, a cursory size-up is performed and rarely is it communicated to other responding units.

Here are **five tips** for a more effective scene size-up to inform other EMS providers and initiate effective incident management.

1. PRACTICE MAKES PERFECT

Standardization is one of the common ways to increase safety and reduce the chance of an error. Many new EMS providers are advised to perform their history and physical exam the same way every time to reduce the chance of missing a portion of the assessment.



Figure 1: What are your scene size-up tips?

<https://media.cdn.lexipol.com/article-images/fb-2ems1dblrsly.jpg?w=300&format=jpg&quality=87&crop=0%2C0%2C420%2C360>

2. COMMUNICATE EARLY, COMMUNICATE OFTEN

As a general rule, EMS providers entered the industry to care for the sick and injured. As a result, on scenes where a size-up is most important, there are likely one or more patients who could really use your help. There is a drive among EMS providers to render aid as quickly as possible and while that goal is admirable it can get you into trouble.

3. SAFETY FIRST

Among the other often-recited phrases in EMS education is "scene safety, BSI." Think about all of the nearly automatic ways EMS providers protect themselves on a daily basis. Crews generally park at an accident scene so that a wayward driver would strike a heavy and unoccupied fire engine before the crews or the ambulance. In the winter, you may back up a patient's driveway so that you don't have to navigate a loaded gurney down a sheet of ice to the ambulance.

4. A WAY IN AND A WAY OUT

Identifying ingress and egress route serves both as a convenience and as a means to ensure provider safety. On a standard call, your ingress route is generally whatever street you happened to be driving down to get to the address and your egress route is simply continuing down that street but with a patient on board. In some cases, however, establishing ingress and egress routes may occur in conjunction with fire and law enforcement colleagues and communicating that to other units could be the difference between a successful or unsuccessful incident.

5. ASK FOR HELP

This tip actually serves two purposes. On scene when performing the size-up, close your initial communication to dispatch with a request for resources based on the estimated number and severity of patients. This means that you may cancel or downgrade responding units later as triage continues. Nothing confuses a request for resources more quickly than asking for "one more ambulance" several times in quick succession. You will lose track of what you've asked for and what is on the way and so will dispatch.

Assessment of nature and extent of injury and /or illness

Assessment of Injuries

The TOTAPS method (Talk, Observe, Touch, Active movement, Passive Movement, Skills Test) is an effective way to assess the nature and extent of an injury to a player and make a decision as to whether or not the athlete who is injured can resume playing.

Talk: The athlete should be asked a range of questions to establish what happened and how they were feeling. Potential questions include: Where is the pain? Did you hear anything? What happened to cause it? Is this the first time this has happened?

Observe: The injured site must be carefully inspected identify any signs of swelling and deformity. It can help to compare the injured site to the other side of the body. (E.g., compare one ankle to the other, to assess an ankle injury). Signs of deformity or swelling are an indication of either soft or hard tissue damage, which means further assessment and treatment from a professional is required.

Touch: If observation fails to yield results the next step involves feeling around the injured site to identify any deformity or swelling. Gentle pressure should be applied from the outside of the injury before slowly working in.

Active Movement: The athlete will need to test movement around the injured site by performing actions such as flexion, extension and rotation to determine whether the mobility of the joint has been affected. If there is no pain whilst the athlete is performing the movement by themselves then they need to complete an isometric contraction before the athlete can move ahead to the last step. If the assessor feels that the athlete does not have a full range of movement, they should progress to the next stage.

Passive Movement: If full range of movement is not evident, the assessor or sports trainer needs to physically mobilize the joint to test the range of motion and identify any instability and painful areas. The athlete must be watched to see if they exhibit signs of discomfort.

Skills Test: The final stage requires the athlete to complete sport specific movements that mimic those which are performed during the game. For example, walking, jogging, side stepping, changing direction and jumping. If the athlete can complete these and the assessor is satisfied, then they can return to play. During this test the assessor must watch the athlete

for any signs of discomfort or favor towards one side. At this stage, strapping may be used to reinforce the injured site.

The athlete must complete and pass all assessment procedures in order to be considered fit enough to return to the playing field.

If the athlete is unable to complete any of the requirements from TOTAPS, then they should be referred to a medical professional for further treatment. The assessment of the injury can be stopped at any time if it is evident that the athlete is feeling pain and cannot return to the field.

Sourcing for emergency services

When an emergency hits, simplified emergency procurement policies and procedures should be enacted to ensure that procurement can be fast enough to meet the emergency needs while also ensuring adequate levels of accountability.

Lessons learned have consistently demonstrated that in the first stages of an emergency, regular organization procurement procedures are overly cumbersome or inadequate to deal with the fast-paced and urgent demands required for emergency procurement. Simplified procedures must be put in place for the emergency period, or considerable slowing of the procurement process will result, which will delay the entire relief effort.

Decisions about what modifications to procedures will be applied in an emergency should ideally be addressed through the EPP process and be able to be activated immediately when an emergency is declared. The most effective organizations are those that already have emergency procedures documented as part of their regular policies, and they simply need to activate them at the time the emergency is declared.

Decision making-Triaging

Patient triage in Accident and Emergency departments requires emergency nurses to make rapid decisions based on their knowledge and experiences. The development of triage decision-making skills can be addressed through the use of simulations, 'thinking aloud' technique, reflection and the decision rules of experienced emergency nurses. Clinical educators and experienced emergency nurse mentors are encouraged to recognize that skill acquisition in triage decision making requires practice before registered nurses can engage fully in the process of triaging patients in the emergency department. It is essential to experience the process of triage decision making in order to develop an understanding of the clinical information attended to, the sequence in which the information is processed and the rules used to combine information leading to a decision on the triage category for each patient. By using triage simulations developed from 'real triage cases' the process of decision making can be experienced by nurses. Further, if these simulations are accompanied by the collection of verbal protocols, nurses have opportunities retrospectively to explore their decision making with reflection. In addition, the presentation and use of decision rules used by experienced triage nurses can enhance the development of skills in novice triage nurses.

Requirements for first aid



Figure 2: Workplace first aid kit

<https://www.ishn.com/ext/resources/todaysnews2/first-aid-kit-422.jpg?t=1424199076&width=696>

IN EXERCISE of the powers conferred by section 55 of the Factories Act, the Minister for Labor after consulting the Labor Advisory Board hereby makes the following Rules:-

THE FACTORIES (FIRST-AID) RULES

In every factory there shall be provided, maintained and readily available and accessible in every first aid box or cupboard required under section 50(1) of the Act items in accordance with the following –

(a) In every factory in which the number of employees does not exceed ten—

- i. a copy of the first-aid leaflet (L.D. 250/1) issued by the Labor Department;
- ii. a sufficient number (not less than six) of small sterilized unmediated dressings for injured fingers;
- iii. a sufficient number (not less than three) of medium sized sterilized unmediated dressing for injured hands or feet;
- iv. a sufficient number (not less than three) of large sterilized unmediated dressings for other injured parts;
- v. a sufficient number (not less than twelve) of adhesive wound dressings of a suitable type and of assorted sizes;
- vi. a sufficient number (not less than two) of triangular bandages of unbleached calico, the longest side of which measures not less than one hundred and thirty centimeters and each of the other sides not less than ninety-one centimeters.;

- vii. a sufficient supply of adhesive plaster;
- viii. a sufficient supply of absorbent sterilized cotton wool in fourteen gramme packets;
- ix. a sufficient supply of Factory Eye Drops, B.P.C.;
- x. a sufficient number (not less two) of sterilized eye pads in separate sealed packets;
- xi. a tourniquet; and a sufficient supply of safety pins;

(b) in every factory in which the number of employees exceeds ten but does not exceed fifty--

- i. a copy of the first-aid leaflet (L.D. 250/1) issued by the Labor Department;
- ii. a sufficient number (not less than twelve) of small sterilized unmediated dressings for injured fingers;
- iii. A sufficient number (not less than six) of medium sized sterilized unmediated dressing for injured hands or feet.
- iv. a sufficient number (not less than six) of large sterilized unmediated dressings for other injured parts;
- v. a sufficient number (not less than twenty four) of adhesive wound dressings of a suitable type and of assorted sizes;
- vi. a sufficient number (not less than four) of triangular bandages of unbleached calico, the longest side of which measures not less than one hundred and thirty centimeters and each of the other sides not less than ninety-one centimeters;
- vii. a sufficient supply of adhesive plaster;
- viii. a sufficient supply of absorbent sterilized cotton wool in fourteen gram packets;
- ix. a sufficient supply of Factory Eye Drops
- x. a sufficient number (not less four) of sterilized eye pads in separate sealed packets;
- xi. a tourniquet; and a sufficient supply of safety pins

(c) in every factory in which the number of employees exceeds fifty--

- i. a copy of the first-aid leaflet (L.D. 250/1) issued by the Labor Department;
- ii. a sufficient number (not less than twenty four) of small sterilized unmediated dressings for injured fingers;
- iii. A sufficient number (not less than twelve) of medium sized sterilized unmediated dressing for injured hands or feet.
- iv. a sufficient number (not less than twelve) of large sterilized unmediated dressings for other injured parts;
- v. a sufficient number (not less than thirty six) of adhesive wound dressings of a suitable type and of assorted sizes;
- vi. a sufficient number (not less than eight) of triangular bandages of unbleached calico, the longest side of which measures not less than one hundred and thirty centimeters and of the other sides not less than ninety-one centimeters;
- vii. a sufficient supply of adhesive plaster;
- viii. a sufficient supply of absorbent sterilized cotton wool in fourteen gram packets;
- ix. a sufficient supply of Factory Eye Drops, B.P.C.;
- x. a sufficient number (not less eight) of sterilized eye pads in separate sealed packets;
- xi. a tourniquet; and a sufficient supply of safety pins;

All materials for drugs and dressings contained in first aid boxes or cupboards shall be those designated in, and of a grade or quality not lower than the standards specified by the British Pharmaceutical Codex or any supplement thereof.

Every first aid box or cupboard shall be plainly and clearly marked on the outside with the words "FIRST AID".

In every factory where there are more than ten but less than fifty employees, there shall be at least two persons trained in first aid and at least one such person shall always be available in the factory at all times during all working hours.

In every factory where there are more than fifty but less than one hundred employees there shall be at least three persons trained in first aid and at least one such person shall always be available in the factory at all times during all working hours.

In every factory where there are more than one hundred employees there shall be at least three persons trained in first aid, plus one additional person for each extra hundred employees (or part thereof) beyond the first 100 employees, and two such trained persons shall always be available in the factory at all times during working hours.

In every factory where more than five hundred persons are employed there shall be a first aid room which shall always be open and manned by a trained nurse during working hours.

No person shall be placed in charge of a first aid box or cupboard unless he or she has received adequate training in the application of first-aid to the injured persons and holds a certificate of competence issued by –

- i. The St. John Ambulance of the St. John Council of Kenya, or
- ii. The Kenya Red Cross Society, or
- iii. Such other body or society as may be approved from time to time, by the Labor Commissioner:

Provided that any certificate of competence issued in accordance with this paragraph shall be valid for one year only and must be renewed annually.

The provisions of these Rules shall apply to both day and night shifts and during all hours a factory is open.

Where offence is committed under these Rules, the owner or the occupier of the factory concerned, as the case may be, shall be guilty of an offence and if the provisions of Part XI of the Act as to offences, penalties (including the amounts thereof) and proceedings shall apply, mutatis mutandis, to these Rules as if they were set out there in.

4.2.1.3 Learning Activities

4.2.1.3.1 Practical Activities

1. Effective assessment of nature and extent of injury or illness is vital in preserving life of the casualties. Using the TOTAPS method demonstrate how you can assess the nature and extent of an injury to a player and make a decision as to whether or not the athlete who is injured can resume playing?

4.2.1.3.2 Self-Assessment

You are provided with the following questions for self-assessment, attempt them and check your responses

1. You will be required to provide first aid services to a variety of patient. What are the principles of first aid?
2. What are the factors you need to consider for a more effective scene size-up to inform other emergency services providers and initiate effective incident management?
3. You are the first to reach to the scene of a patient involved in trauma. Primary survey in the management of trauma involves?
4. Sourcing for emergency services materials and equipment is very vital in the provision of emergency services. What is the importance of effective sourcing for emergency services materials and equipment?
5. Patient triage in Accident and Emergency departments requires emergency nurses to make rapid decisions based on their knowledge and experiences. Why is effective rapid decision making by nurses working in the patient triage in accident and emergency departments very important in saving the lives of the casualties?
6. According to the factories first aid rules, every factory in which the number of employees does not exceed ten should fulfill which requirements?
7. According to the factories first aid rules, every factory in which the number of employees exceeds ten but does not exceed fifty should fulfill which requirements?

4.2.1.4 Tools, Equipment, Supplies and Materials

- Casualty department
- Functional skills laboratory
- First aid kit
- Emergency room
- Emergency room standard manuals
- Emergency trolley
- First Aid Manual.
- Tweezers
- Alcohol Swabs
- Antibiotic Ointment
- Bandages
- Gauze Pads
- Medical Tape
- Elastic Bandages
- Pain relievers
- Instant cold pack
- Waste disposal bins
- Personal protective equipment
- Health workers
- Presentation charts
- Flip charts

4.2.1.5 References

Jevon, P. & Ewens, B. (2012) *Monitoring the Critically Ill Patient*, 3rd Edition, Wiley-Blackwell, Oxford

Peitzman A.B., et al. (2012) *The Trauma Manual*, 2nd edition. Philadelphia, Pa., Lippincott Williams & Wilkins.

Rapid Response to Everyday Emergencies (2016) *A Nurse's Guide*. Philadelphia, Pa., Lippincott Williams & Wilkins.

Royal College of nursing (2017) *Record Keeping: The Facts*, RCN, viewed 28 May 2018, <https://www.rcn.org.uk/professional-development/publications/pub-006051>

MODEL ANSWERS

Self-Assessment

1. You will be required to provide first aid services to a variety of patient. What are the principles of first aid?

- **Preserve Life**--The first aim of first aid is to preserve life, which involves the key emergency practices to ensure that the casualty isn't in any mortal danger.
- **Prevent Deterioration**--Once you've followed all the steps associated with the first principle, your next priority is to prevent deterioration of the injured person's condition. Keeping a casualty still to avoid aggravating their injury, or from complicating any unseen issues, is crucial.
- **Promote Recovery**--Finally, there are steps you should follow which will help lessen the amount of time taken for a casualty to recover from an accident and aid in minimizing lasting damage and scarring.

2. What are the factors you need to consider for a more effective scene size-up to inform other emergency services providers and initiate effective incident management?

- **Practice makes perfect**--Standardization is one of the common ways to increase safety and reduce the chance of an error. Many new emergency service providers are advised to perform their history and physical exam the same way every time to reduce the chance of missing a portion of the assessment.
- **Communicate early, communicate often**—Early and often communication by the emergency service providers about the scene size up help to secure the emergency scene and reduce significantly morbidity and mortality rate of the casualties.
- **Safety first**-- Among the other often-recited phrases in emergency service education is "scene safety." Think about all of the nearly automatic ways emergency service providers protect themselves on a daily basis.
- **A way in and a way out**-- Identifying ingress and egress route serves both as a convenience and as a means to ensure provider safety. On a standard call, your ingress route is generally whatever street you happened to be driving down to get to the address and your egress route is simply continuing down that street but with a patient on board.
- **Ask for help**-- This tip actually serves two purposes. On scene when performing the size-up, close your initial communication to dispatch with a request for resources based on the estimated number and severity of patients. This means that you may cancel or downgrade responding units later as triage continues. Nothing confuses a request for resources more quickly than asking for "one more ambulance" several times in quick succession. You will lose track of what you've asked for and what is on the way and so will dispatch.

3. You are the first to reach to the scene of a patient involved in trauma. Primary survey in the management of trauma involves?

- Airway-- open the patient's airway using a jaw-thrust maneuver with manual, in-line stabilization of the neck. If you find food, blood, vomitus, or other debris, suction the airway quickly to prevent aspiration. To better remove secretions, you may need to carefully logroll the patient to his side. Manually stabilize his neck and spine as you do so.
- Breathing-- Note respiratory rate and depth, chest expansion, and accessory muscle use and auscultate breath sounds bilaterally. Also palpate for crepitus or subcutaneous air in the neck and chest, which can indicate a pneumothorax or airway injury.
- Circulation--Once you've assessed and supported your patient's breathing, attend to his circulatory status. Assess for the presence and quality of peripheral pulses to quickly estimate blood pressure
- Disability--To evaluate disability, you'll evaluate the patient's LOC, pupil response, and gross sensorimotor function. To document his baseline LOC, quickly assess and record an initial GCS score.
- Exposure--The final component of the primary survey is *exposure*. Remove the patient's clothing completely so you can inspect his entire body for injuries. Use good judgment when removing clothing; trying to remove a shirt by pulling or manipulating it may worsen the injury or pain. Cutting clothing away with trauma shears is usually best.

4. Sourcing for emergency services materials and equipment is very vital in the provision of emergency services. What is the importance of effective sourcing for emergency services materials and equipment?

- When an emergency hits, simplified emergency procurement policies and procedures should be enacted to ensure that procurement can be fast enough to meet the emergency needs while also ensuring adequate levels of accountability.
- Lessons learned have consistently demonstrated that in the first stages of an emergency, regular organization procurement procedures are overly cumbersome or inadequate to deal with the fast-paced and urgent demands required for emergency procurement. Simplified procedures must be put in place for the emergency period, or considerable slowing of the procurement process will result, which will delay the entire relief effort.
- Decisions about what modifications to procedures will be applied in an emergency should ideally be addressed through the emergency procurement procedures process and be able to be activated immediately when an emergency is declared. The most effective organizations are those that already have emergency procedures documented as part of their regular policies, and they simply need to activate them at the time the emergency is declared.

5. Patient triage in Accident and Emergency departments requires emergency nurses to make rapid decisions based on their knowledge and experiences. Why is effective rapid decision making by nurses working in the patient triage in accident and emergency departments very important in saving the lives of the casualties?

- The development of triage decision-making skills can be addressed through the use of simulations, 'thinking aloud' technique, reflection and the decision rules of experienced emergency nurses.
- Clinical educators and experienced emergency nurse mentors are encouraged to recognize that skill acquisition in triage decision making requires practice before registered nurses can engage fully in the process of triaging patients in the emergency department.
- It is essential to experience the process of triage decision making in order to develop an understanding of the clinical information attended to, the sequence in which the information is processed and the rules used to combine information leading to a decision on the triage category for each patient.
- By using triage simulations developed from 'real triage cases' the process of decision making can be experienced by nurses.
- Further, if these simulations are accompanied by the collection of verbal protocols, nurses have opportunities retrospectively to explore their decision making with reflection.
- In addition, the presentation and use of decision rules used by experienced triage nurses can enhance the development of skills in novice triage nurses.

6. According to the factories first aid rules, every factory in which the number of employees does not exceed ten should fulfill which requirements?

- a copy of the first-aid leaflet (L.D. 250/1) issued by the Labor Department;
- a sufficient number (not less than six) of small sterilized unmediated dressings for injured fingers;
- a sufficient number (not less than three) of medium sized sterilized unmediated dressing for injured hands or feet;
- a sufficient number (not less than three) of large sterilized unmediated dressings for other injured parts;
- a sufficient number (not less than twelve) of adhesive wound dressings of a suitable type and of assorted sizes;
- a sufficient number (not less than two) of triangular bandages of unbleached calico, the longest side of which measures not less than one hundred and thirty centimeters and each of the other sides not less than ninety-one centimeters.;
- a sufficient supply of adhesive plaster;
- a sufficient supply of absorbent sterilized cotton wool in fourteen gramme packets;
- a sufficient supply of Factory Eye Drops, B.P.C.;
- a sufficient number (not less two) of sterilized eye pads in separate sealed packets;
- a tourniquet; and a sufficient supply of safety pins;

7. According to the factories first aid rules, every factory in which the number of employees exceeds ten but does not exceed fifty should fulfill which requirements?

- a copy of the first-aid leaflet (L.D. 250/1) issued by the Labor Department;
- a sufficient number (not less than twelve) of small sterilized unmediated dressings for injured fingers;

- A sufficient number (not less than six) of medium sized sterilized unmediated dressing for injured hands or feet.
- a sufficient number (not less than six) of large sterilized unmediated dressings for other injured parts;
- a sufficient number (not less than twenty four) of adhesive wound dressings of a suitable type and of assorted sizes;
- a sufficient number (not less than four) of triangular bandages of unbleached calico, the longest side of which measures not less than one hundred and thirty centimeters and each of the other sides not less than ninety-one centimeters;
- a sufficient supply of adhesive plaster;
- a sufficient supply of absorbent sterilized cotton wool in fourteen gram packets;
- a sufficient supply of Factory Eye Drops
- a sufficient number (not less four) of sterilized eye pads in separate sealed packets;
- a tourniquet; and a sufficient supply of safety pins

4.2.1 Learning Outcome 2:

Provide first aid intervention

4.2.1.1 Introduction to the learning outcome

This learning outcome specifies the content of competencies required to provide first aid interventions. It includes performing cardio- pulmonary resuscitation; management of: choking, burns and scalds, trauma, nose bleeding, cuts, fractures, drowning, poisoning and snake bites, assisting patients with medical conditions such as: asthma, epilepsy and diabetes, effective communication on casualty's condition, handing over casualty to medical personnel and monitoring Casualty's condition.

4.2.1.2 Performance Standard

1. Gave specific first aid measures based on nature and extent of injury/illness
2. Monitored casualty's condition continuously based on the vital signs
3. Conveyed information on the casualty condition accurately and clearly to emergency services personnel
4. Handed over the casualty to the emergency and appropriate medical personnel

4.2.1.3 Information Sheet

Definition of Terms

Cardio- pulmonary resuscitation- Cardiopulmonary resuscitation (CPR) is an emergency procedure that combines chest compressions often with artificial ventilation in an effort to manually preserve intact brain function until further measures are taken to restore spontaneous blood circulation and breathing in a person who is in cardiac arrest.

Casualty- the department of a hospital providing immediate treatment for emergency cases

First Aid- is the first and immediate assistance given to any person suffering from either a minor or serious illness or injury, with care provided to preserve life, prevent the condition from worsening, or to promote recovery

Emergency services- are the public organizations whose job is to take quick action to deal with emergencies when they occur, especially the fire brigade, the police, and the ambulance service.

Injury- also known as physical trauma is damage to the body caused by external force. This may be caused by accidents, falls, hits, weapons, and other causes.

Performing cardio- pulmonary resuscitation

Using the CPR steps on someone who is not breathing can help keep them alive until the emergency services arrive.

CPR works by keeping a person's blood flowing until healthcare professionals can help them. People without first aid training can still save a life by using the CPR steps.

When a person initiates CPR immediately after someone's heart stops beating, CPR can double or even triple the chances of them surviving.

CPR step-by-step

There are two main stages to CPR: the preparation stage and the CPR stage.

Preparation steps

Before performing CPR on an adult, use the following preparation steps:

Step 1 Call 911

First, check the scene for factors that could put you in danger, such as traffic, fire, or falling masonry. Next, check the person. Do they need help? Tap their shoulder and shout, "Are you OK?"

If they are not responding, call 911 or ask a bystander to call 911 before performing CPR. If possible, ask a bystander to go and search for an AED machine. People can find these in offices and many other public buildings.

Step 2 Place the person on their back and open their airway

Place the person carefully on their back and kneel beside their chest. Tilt their head back slightly by lifting their chin.

Open their mouth and check for any obstruction, such as food or vomit. Remove any obstruction if it is loose. If it is not loose, trying to grasp it may push it farther into the airway.

Step 3 Check for breathing

Place your ear next the person's mouth and listen for no more than 10 seconds. If you do not hear breathing, or you only hear occasional gasps, begin CPR.

If someone is unconscious but still breathing, do not perform CPR. Instead, if they do not seem to have a spinal injury, place them in the recovery position. Keep monitoring their breathing and perform CPR if they stop breathing.

CPR steps

Use the following steps to perform CPR:

Step 4 Perform 30 chest compressions

Place one of your hands on top of the other and clasp them together. With the heel of the hands and straight elbows, push hard and fast in the center of the chest, slightly below the nipples.

Push at least 2 inches deep. Compress their chest at a rate of at least 100 times per minute. Let the chest rise fully between compressions.

Chest compressions



Adult



press down
↓ **2 inches**



Child



press down
↓ **2 inches**



Infant



press down
↓ **1.5 inches**

Perform 30 chest compressions at a rate of 100 per minute, letting the chest rise between each

Figure 3: Chest compressions techniques

<https://cdn-prod.medicalnewstoday.com/content/images/articles/324/324712/chest-compressions-in-cpr-visual-guide-infographic.jpg>

Step 5 Perform two rescue breaths

Making sure their mouth is clear, tilt their head back slightly and lift their chin. Pinch their nose shut, place your mouth fully over theirs, and blow to make their chest rise.

If their chest does not rise with the first breath, retilt their head. If their chest still does not rise with a second breath, the person might be choking.

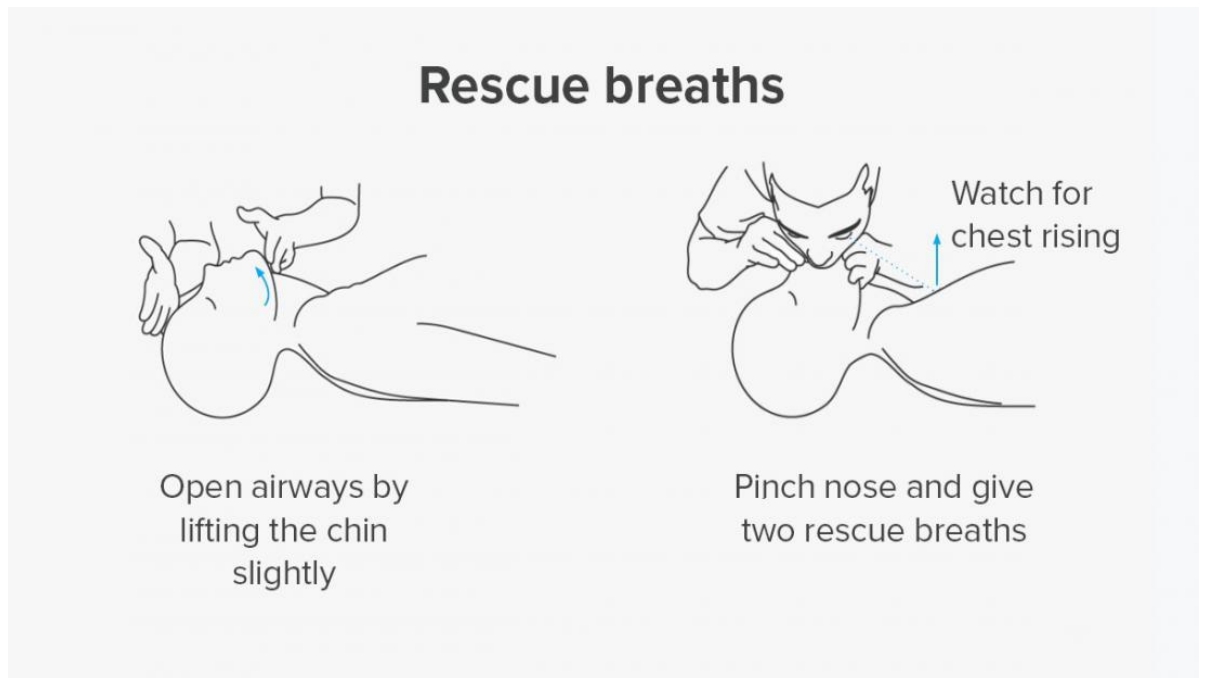


Figure 4: Rescue breaths

<https://cdn-prod.medicalnewstoday.com/content/images/articles/324/324712/cpr-rescue-breaths-visual-guide-illustration.jpg>

Step 6 Repeat

Repeat the cycle of 30 chest compressions and two rescue breaths until the person starts breathing or help arrives. If an AED arrives, carry on performing CPR until the machine is set up and ready to use.



Figure 5: How to Do CPR - Step by Step Cardiopulmonary Resuscitation

<https://www.trainingexpress.org.uk/wp-content/uploads/2020/02/How-to-Do-CPR-Step-1-e1581920675299.jpg>

Management of: Choking

Choking happens when someone's airway suddenly gets blocked, either fully or partly, so they can't breathe.

This information applies to adults and children over 1 year old.

Mild choking: encourage them to cough

If the airway is only partly blocked, the person will usually be able to speak, cry, cough or breathe.

They'll usually be able to clear the blockage themselves.

To help with mild choking in an adult or child over 1 year old:

- encourage them to keep coughing to try to clear the blockage
- ask them to try to spit out the object if it's in their mouth

- don't put your fingers in their mouth to help them as they may bite you accidentally
- If coughing doesn't work, start back blows.

Severe choking: back blows and abdominal thrusts

Where choking is severe, the person won't be able to speak, cry, cough or breathe. Without help, they'll eventually become unconscious.

To carry out a back blow on an adult or child over 1 year old:

- Stand behind them and slightly to one side. Support their chest with 1 hand. Lean them forward so the object blocking their airway will come out of their mouth, rather than moving further down.
- Give up to 5 sharp blows between their shoulder blades with the heel of your hand. The heel is between the palm of your hand and your wrist.
- Check if the blockage has cleared.
- If not, give up to 5 abdominal thrusts.

Abdominal thrusts

Don't give abdominal thrusts to babies under 1 year old or pregnant women.

To carry out an abdominal thrust:

- Stand behind the person who's choking.
- Place your arms around their waist and bend them forward.
- Clench 1 fist and place it right above their belly button.
- Put the other hand on top of your fist and pull sharply inwards and upwards.
- Repeat this movement up to 5 times.

If the person's airway is still blocked after trying back blows and abdominal thrusts, get help immediately:

- Call 999 and ask for an ambulance. Tell the 999 operator the person is choking.
- Continue with the cycles of 5 back blows and 5 abdominal thrusts until help arrives.

If they lose consciousness and aren't breathing, you should begin cardiopulmonary resuscitation (CPR) with chest compressions.

Complications of Choking

Get urgent medical help at an A&E, NHS walk-in centre or a GP if:

- they have a persistent cough after choking

- they feel something is still stuck in their throat

Abdominal thrusts can cause serious injuries. A health professional such as your GP or a doctor in A&E should always examine someone after they have received abdominal thrusts.

Management of Burns and Scalds

About burns and scalds



Figure 6: Burns and Scalds

<https://assets.nhs.uk/nhsuk-cms/images/C0034381.width-320.jpg>

Burns and scalds are damage to the skin caused by heat. Both are treated in the same way.

A burn is caused by dry heat – by an iron or fire, for example. A scald is caused by something wet, such as hot water or steam.

Burns can be very painful and may cause:

- red or peeling skin
- blisters
- swelling
- white or charred skin

The amount of pain you feel isn't always related to how serious the burn is. Even a very serious burn may be relatively painless.

Treating burns and scalds

To treat a burn, follow the first aid advice below:

- immediately get the person away from the heat source to stop the burning
- cool the burn with cool or lukewarm running water for 20 minutes – don't use ice, iced water, or any creams or greasy substances such as butter
- remove any clothing or jewellery that's near the burnt area of skin, including babies' nappies - but don't move anything that's stuck to the skin
- make sure the person keeps warm – by using a blanket, for example, but take care not to rub it against the burnt area
- cover the burn by placing a layer of cling film over it – a clean plastic bag could also be used for burns on your hand
- use painkillers such as paracetamol or ibuprofen to treat any pain
- if the face or eyes are burnt, sit up as much as possible, rather than lying down - this helps to reduce swelling

The British Red Cross website has a [video about first aid for burns](#).

When to get medical attention

Depending on how serious a burn is, it may be possible to treat it at home. For minor burns, keep the burn clean and don't burst any blisters that form.

More serious burns require professional medical attention. You should go to a hospital A&E department for:

- all chemical and electrical burns
- large or deep burns – any burn bigger than your hand
- burns that cause white or charred skin – any size
- burns on the face, hands, arms, feet, legs or genitals that cause blisters

If someone has breathed in smoke or fumes, they should also seek medical attention. Some symptoms may be delayed and can include:

- coughing
- a sore throat
- difficulty breathing
- facial burns

People at greater risk from the effects of burns, such as children under five years old and pregnant women, should also get medical attention after a burn or scald.

The size and depth of the burn will be assessed and the affected area cleaned before a dressing is applied. In severe cases, skin graft surgery may be recommended.

Types of burn

Burns are assessed by how seriously your skin is damaged and which layers of skin are affected. Your skin has three layers:

- the epidermis – the outer layer of skin
- the dermis – the layer of tissue just beneath, which contains blood capillaries, nerve endings, sweat glands and hair follicles
- the subcutaneous fat, or subcutis – the deeper layer of fat and tissue

There are four main types of burn, which tend to have a different appearance and different symptoms:

- superficial epidermal burn – where the epidermis is damaged; your skin will be red, slightly swollen and painful, but not blistered
- superficial dermal burn – where the epidermis and part of the dermis are damaged; your skin will be pale pink and painful, and there may be small blisters
- deep dermal or partial thickness burn – where the epidermis and the dermis are damaged: this type of burn makes your skin turn red and blotchy; your skin may be dry or moist, and become swollen and blistered, and it may be very painful or painless
- full thickness burn – where all three layers of skin (the epidermis, dermis and subcutis) are damaged; the skin is often burnt away and the tissue underneath may appear pale or blackened, while the remaining skin will be dry and white, brown or black with no blisters, and the texture of the skin may also be leathery or waxy

Preventing burns and scalds

Many severe burns and scalds affect babies and young children. Examples of things you can do to help reduce the likelihood of your child having a serious accident at home include:

- keeping your child out of the kitchen whenever possible
- testing the temperature of bath water using your elbow before you put your baby or toddler in the bath
- keeping matches, lighters and lit candles out of young children's sight and reach
- keeping hot drinks well away from young children

Management of Trauma

Primary Survey: Managing immediate threats

By taking a standardized approach to assessment and treatment, the trauma team can address the most significant risks to life first. As always, start with the ABCs.

Airway. The first part of the primary survey is always assessing the airway. This includes checking for potential injury to the cervical spine. Until cervical spine injury has been ruled out, open the patient's airway using a jaw-thrust maneuver with manual, in-line stabilization of the neck. If you find food, blood, vomitus, or other debris, suction the airway quickly to

prevent aspiration. To better remove secretions, you may need to carefully logroll the patient to his side. Manually stabilize his neck and spine as you do so.

Breathing. Assess your patient's breathing next. Note respiratory rate and depth, chest expansion, and accessory muscle use and auscultate breath sounds bilaterally. Also palpate for crepitus or subcutaneous air in the neck and chest, which can indicate a pneumothorax or airway injury. Find out if he has pain with breathing or on palpation. Injuries that can impair ventilation include rib fractures (especially a flail chest), a pneumothorax, a hemothorax, and spinal cord or head trauma.

Supplemental oxygen is always indicated at this stage. For a spontaneously breathing patient, a non-rebreather mask with the flow rate set at 12 to 15 liters/minute is appropriate. However, if the patient isn't breathing well enough to sustain optimal oxygenation, begin manual bag-valve-mask ventilation to support his ventilatory efforts until he can be intubated and mechanically ventilated.

If the patient has severe respiratory distress and hypotension as well as unilateral decreased or absent breath sounds, suspect a tension pneumothorax, a potentially fatal complication requiring rapid treatment. To perform an emergency chest decompression, the trauma team physician will perform a needle thoracostomy, inserting a 14-gauge I.V. catheter into the patient's chest at the second intercostal space, midclavicular line on the affected side. A rush of air from the catheter confirms the presence of a tension pneumothorax. The catheter is left in place until a chest tube can be inserted.

In the meantime, a syringe or commercial Heimlich valve (or similar device) is attached to the catheter hub so that air can escape without being drawn back into the chest. If available, have a chest tube drainage system that can collect blood for autotransfusion on hand during chest tube insertion, in case a hemothorax is present.

Circulation. Once you've assessed and supported your patient's breathing, attend to his circulatory status. Assess for the presence and quality of peripheral pulses to quickly estimate BP, as follows.

- If he has a radial pulse, his systolic BP is at least 80 mm Hg.
- If he's lost his radial pulse but still has a femoral pulse, he has a systolic BP of at least 70 mm Hg.
- If he lacks all pulses except a carotid pulse, he has a systolic BP of at least 60 mm Hg.

Note the patient's skin color and level of consciousness (LOC). Pallor and cold, clammy skin indicate shock.

His LOC is an important indicator of cerebral perfusion. Agitation is common in the early stages of shock. (Think of the "fight or flight" response.) As shock progresses, his LOC will decline until he's unconscious.

Obtain a complete set of vital signs, including temperature, as soon as possible. Use this set of vital signs as a baseline for comparison with subsequent measurements. You may need to take vital signs every 5 to 15 minutes until the patient's condition improves.

A key part of your circulatory assessment is to identify and control hemorrhage. *External hemorrhage* is usually, but not always, obvious. Logroll the patient to inspect his back and buttocks for bleeding.

To control bleeding, apply direct pressure over the site of hemorrhage. If this isn't effective by itself, apply pressure over the major arterial pulse point proximal to the bleeding site.

Use a tourniquet only if you must stanch severe hemorrhage in an extremity to save the patient's life. Using a tourniquet puts the limb's viability at risk.

Next, ask yourself if the mechanism of injury makes *internal hemorrhage* likely. If the patient has signs and symptoms of shock without visible bleeding, he may have an occult internal hemorrhage that requires surgery.

Besides assessing and documenting his circulatory status, you may need to intervene to sustain circulation. For a patient who's in shock, consider both noninvasive and invasive strategies to support his BP. Keep him supine and elevate his legs 6 to 8 inches (15 to 20 cm) to promote venous return and improve cardiac output. Don't put him in the Trendelenburg position because this can cause his stomach to compress his diaphragm, impairing ventilation.

Make sure he has venous access with two large-bore I.V. catheters (ideally 14- to 16-gauge) to facilitate rapid fluid and blood product administration if needed. Draw blood for lab analysis. Send specimens for typing and crossmatching, complete blood cell count, serum glucose, electrolytes, and a coagulation profile. Depending on the patient's condition and suspected injuries, you may also need specimens for other studies, such as creatine kinase, amylase, and serum lactate.

An arterial blood gas (ABG) analysis can help clinicians assess the patient's oxygenation status and determine whether or not he's in shock. If ABG results show a base deficit that's greater than 2 mEq/liter, suspect ongoing hemorrhage, internal injuries, or insufficient resuscitation.

As ordered, administer an appropriate crystalloid solution for I.V. volume replacement, such as 0.9% sodium chloride or lactated Ringer's solution. Warm the solution in a commercial fluid warmer or use a high-volume infuser/warming device. Don't administer D₅W for volume replacement because the dextrose will be metabolized and leave free water, a hypotonic solution that won't stay in the vascular space.

Provide 3 mL of crystalloid solution to replace each 1 mL of blood lost. If you infuse 2 liters of crystalloid solution and the patient's BP hasn't returned to the normal range, be prepared to administer blood products.

Typing and crossmatching typically take 30 to 40 minutes, which may be too long for a trauma patient to wait. When immediate blood transfusion is needed, the only option is to give uncrossmatched universal donor blood, as ordered. Give group O, Rh-negative packed red blood cells (RBCs) to female patients of childbearing age or younger. Male patients and

women who can't become pregnant can receive group O, Rh-positive blood. Remember that 0.9% sodium chloride is the only solution you can infuse in the same I.V. line as blood.

Disability--To evaluate disability, you'll evaluate the patient's LOC, pupil response, and gross sensorimotor function. To document his baseline LOC, quickly assess and record an initial GCS score. If possible, determine his GCS before he receives any drugs that could alter his LOC to better enable you to predict his outcome. For example, if a patient's GCS score on arrival at the hospital is 4, his prognosis for recovery is much worse than a patient whose initial score is 12.

Keep in mind that accurate scoring can be impaired by traumatic, toxic, and metabolic causes. Even if the patient shows evidence of alcohol or drug use, never assume that his altered mental status is due purely to intoxicants until injury and other medical causes are ruled out. (See *Using the Glasgow Coma Scale*.)

Note whether the patient can recall the events surrounding the traumatic event. Amnesia about the event suggests that he lost consciousness.

Next, assess his pupils for size, equality, shape, and response to light. Unequal or abnormal pupil response can indicate direct ocular trauma or head injury and elevated intracranial pressure or the effects of drugs, such as atropine (pupil dilation) or opioids (pupil constriction).

The final component of the disability evaluation is an assessment of gross sensorimotor function. Try to determine if the patient has any numbness, tingling, or other abnormal sensations in his body after the traumatic event and if he can move his limbs. Injuries to the extremities, spinal cord, head, blood vessels, or nerves can cause sensorimotor deficits.

Exposure--The final component of the primary survey is *exposure*. Remove the patient's clothing completely so you can inspect his entire body for injuries. Use good judgment when removing clothing; trying to remove a shirt by pulling or manipulating it may worsen the injury or pain. Cutting clothing away with trauma shears is usually best.

Once you've removed clothing, protect the patient from hypothermia, which is particularly dangerous to any trauma patient because it impairs blood coagulation, interferes with resuscitation efforts, and increases the risk of acidosis and death.

Take these measures to prevent heat loss and rewarm the patient.

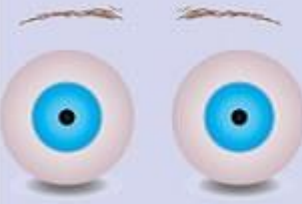
Behaviour	Response
 Eye Opening Response	4. Spontaneously 3. To speech 2. To pain 1. No response
 Verbal Response	5. Oriented to time, person and place 4. Confused 3. Inappropriate words 2. Incomprehensible sounds 1. No response
 Motor Response	6. Obeys command 5. Moves to localised pain 4. Flex to withdraw from pain 3. Abnormal flexion 2. Abnormal extension 1. No response

Table 1: The Glasgow Coma Scale

<https://biologydictionary.net/wp-content/uploads/2020/10/gcs.jpg>

*Remove wet clothing and sheets. Cover the patient with warm blankets.

* Increase the room temperature to 75[degrees] F to 80[degrees] F (23.9[degrees] C to 26.7[degrees] C).

* Infuse only warm crystalloid solutions.

* Consider using commercial patient-warming devices, such as heat lights or temperature-regulating blankets.

Secondary survey: Uncovering other serious threats

Once you've completed the primary survey and managed any immediate threats to the patient's life, begin a secondary survey for injuries that could kill or disable him later. Start at his head and assess him methodically, moving down his body systematically as you search for injuries. Inspect for contusions, abrasions, lacerations, deformities, discoloration, edema, foreign bodies, and other abnormalities.

Auscultate breath sounds and heart sounds. Assess all body areas to locate areas of pain or tenderness, crepitus, deformity, loss of function, and the location and quality of pulses. If you suspect he has a fracture of an arm or leg, assess the neurovascular status of the limb, then splint it to prevent movement and decrease pain. Assess neurovascular status again after splinting. Administer I.V. opioid analgesia as ordered and make sure that pain is managed optimally.

At this point, the trauma physician will consider ordering an indwelling urinary catheter to accurately measure urinary output, an indication of renal perfusion, and to check for blood in the urine. First, though, he'll perform a rectal examination to check for blood or evidence of urethral injury, such as a high-riding prostate gland in a male patient. (If the urethra is injured, the patient may need to have a suprapubic catheter inserted instead.)

Before inserting a urinary catheter, look for blood at the urethral meatus. If you see blood, notify the physician and don't insert the catheter. The patient will need further diagnostic testing (for instance, a retrograde urethrogram or cystogram) before a catheter can be safely inserted.

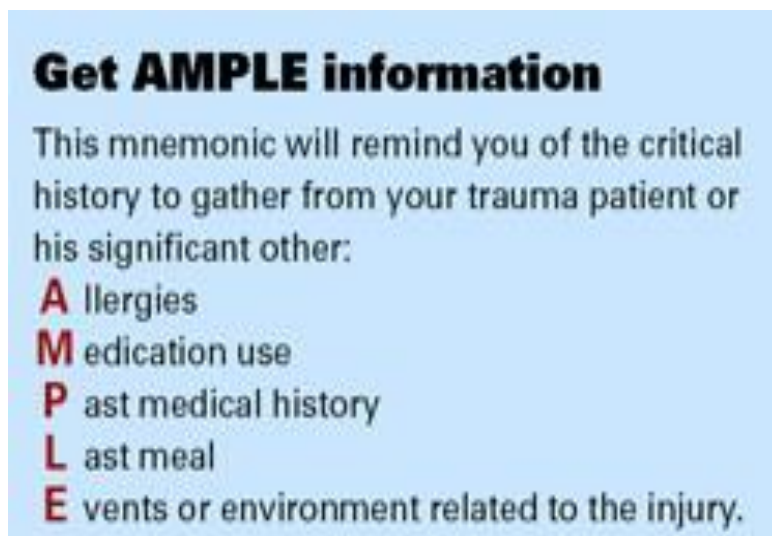


Figure 7: Responding to trauma: Your priorities in the first hour

https://www.nursingcenter.com/wkhlp/handlers/Cv.png?key=article_TTU2_01244666-200901000-00010

Reassess the patient's vital signs and GCS score as frequently as needed, depending on his condition. Also try to obtain a more complete history from the patient or significant others. Use the mnemonic "AMPLE" to help you remember the key information to gather. (See *Get AMPLE information.*)

Assess carefully for medications the patient has taken that could affect his condition and treatment. For example, taking an anticoagulant, such as warfarin, or a platelet inhibitor, such as daily aspirin therapy, will make him much more prone to bleeding from his injuries. If he's using any of these drugs, tell the health care provider immediately so that he can order appropriate reversal agents or take measures to counteract anticoagulation effects.

Assess the patient for steroid use. If he's taking a steroid medication, he may need an I.V. steroid bolus so that he can physiologically respond in a stress or shock state. If you don't know the date of his last tetanus immunization or if it was more than 5 years ago, administer tetanus prophylaxis.

Michael's secondary survey is remarkable for pain on palpation in his lumbar spine, tenderness and abrasions over his left lower rib cage anteriorly, and heel pain and swelling in both feet. You insert a urinary catheter and perform a dipstick urine test, which is positive for a small amount of blood.

Next up: An eye on Diagnostics

After the primary and secondary surveys are complete, prepare your patient for a series of X-rays and scans. He'll have a stat portable chest X-ray to identify rib fractures or mediastinal or diaphragmatic injury and to assess for a pneumothorax or hemothorax. He'll also need a cervical spine X-ray series to check for cervical spine injury. The X-ray will also confirm the correct position of chest and endotracheal tubes and central venous catheters. Depending on the results of the primary and secondary surveys, he may have additional X-rays of the pelvis, spine, extremities, or other areas.

He may have bedside ultrasonography with the focused abdominal sonography for trauma (FAST) technique, which is used to rapidly examine all four abdominal quadrants and the pericardium to identify the presence of free fluid, usually blood.

If he's lost consciousness or shows evidence of a head injury, he'll need a computed tomography (CT) scan of his head. Other CT scans of the spine, chest, abdomen, or pelvis may be indicated to help the health care provider plan treatment.

Your patient may need a vascular ultrasound or an arteriogram if he has vascular injuries, decreased or absent pulses, evidence of limb ischemia, or a widened mediastinum, indicating a possible aortic injury.

Magnetic resonance imaging (MRI) is rarely used for diagnosing acutely injured patients because it takes too long and safely placing an injured patient into the MRI tube is difficult. In addition, the patient might have ferrous metal in his body (for example, implants, or metal fragments left in his eyes from industrial work). Any ferrous metal is dangerous in an MRI room and is a contraindication for MRI.

However, the patient may need an MRI if he shows any evidence of an acute spinal cord injury. Be sure to carefully assess him for ferrous metal objects. If they can be removed, do so before taking him to the MRI. The technologist will ask him if he has any implants or fragments in his eyes from metal work. If he does, an MRI is contraindicated.

Michael's diagnostic workup includes a bedside FAST examination; chest, pelvis, and lower extremity X-rays; a full series of spinal X-rays; and CT scans of his chest, abdomen, and lumbar spine. The tests identify these injuries: fractures of the 9th and 10th ribs on the left side, an L3 compression fracture, bilateral calcaneus fractures, a renal contusion, and a grade III spleen injury.

Providing Definitive Care

The definitive care phase begins after the patient's injuries have been identified and initial lifesaving interventions have been performed. If your hospital doesn't have the resources to provide the care he needs, he may need to be transferred to a trauma center.

In a facility that can provide trauma management, the patient may go to the operating room, intensive care unit (ICU), or a surgical unit after his trauma workup. Most patients go home after discharge, but some require inpatient rehabilitation first.

In Michael's case, the surgeon admits him to the ICU for close monitoring and pain management. She elects to manage his spleen injury nonoperatively because his vital signs normalized after he received 2 liters of resuscitation fluids. His rib fractures and renal contusion require only observation at this time. Orthopedic and spine surgeons are consulted to treat his calcaneus fractures and L3 compression fracture.

Meeting the Standard of Care

Key outcome measures will help you to determine how well the patient has responded to resuscitation and help you anticipate his needs. (See *Adequate resuscitation? Watch for these indicators*.)

An organized team approach in the first hour after a traumatic injury provides fast, efficient patient care and saves lives. Because you and other team members prioritized assessment and interventions for Michael according to recognized standards of trauma care, you've given him the best chance for survival and a full recovery.

Adequate Resuscitation? Watch for these indicators

- Hemodynamic and renal parameters within normal limits
- Core body temperature normal
- Serum lactate less than 2 mmol/liter
- No base deficit
- Arterial pH of 7.35 to 7.45
- Hemoglobin greater than 9 grams/dL (based on individual needs)

- Ionized calcium within normal limits. (Blood transfusion can lower serum calcium because of the calcium-binding effects of the citrate preservative in banked blood products.)
- Serum potassium of 3.5 to 5.3 mEq/liter
- Coagulation profile within normal limits
- Pain under control

Management of Nose Bleeding

About nosebleeds



Figure 8: Nose Bleeds First Aid

<https://ebgh.med.sa/wpcontent/uploads/2017/10/%D8%A7%D9%84%D8%B1%D8%B9%D8%A7%D9%81-400x280.jpg?i=56491>

Nosebleeds can be frightening, but they aren't usually a sign of anything serious and can often be treated at home.

The medical name for a nosebleed is epistaxis.

During a nosebleed, blood flows from one or both nostrils. It can be heavy or light and last from a few seconds to 15 minutes or more.

What to do

To stop a nosebleed:

- sit down and firmly pinch the soft part of your nose, just above your nostrils, for at least 10-15 minutes
- lean forward and breathe through your mouth – this will drain blood into your nose instead of down the back of your throat
- place an ice pack or bag of frozen vegetables covered by a towel on the bridge of your nose
- stay upright, rather than lying down, as this reduces the blood pressure in the blood vessels of your nose and will discourage further bleeding

If the bleeding eventually stops, you won't usually need to seek medical advice. However, in some cases you may need further treatment from your doctor or in hospital (see below).

Read more about [treating nosebleeds](#)

When to seek medical advice

Contact your Doctor if:

- you're taking a blood-thinning medicine (anticoagulant) such as warfarin or have a clotting disorder such as haemophilia and the bleeding doesn't stop
- you have symptoms of [anaemia](#) such as heart palpitations, [shortness of breath](#) and a pale complexion
- a child under two years of age has a nosebleed
- you have nosebleeds that come and go regularly

Ask someone to drive you to your nearest accident and emergency (A&E) department or call 999 for an ambulance if:

- the bleeding continues for longer than 20 minutes
- the bleeding is heavy and you've lost a lot of blood
- you're having difficulty breathing
- you swallow a large amount of blood that makes you vomit
- the nosebleed developed after a serious injury, such as a car crash

What causes nosebleeds?

The inside of your nose is full of tiny, delicate blood vessels that can become damaged and bleed relatively easily.

Common causes of nosebleeds include:

- picking your nose
- blowing your nose very hard
- a minor injury to your nose
- changes in humidity or temperature causing the inside of the nose to become dry and cracked

Occasionally, bleeding can come from the blood vessels deeper within the nose. This can be caused by a blow to the head, recent nasal surgery and hardened arteries (atherosclerosis).

Read more about the causes of nosebleeds

Who gets nosebleeds?

Nosebleeds are fairly common and most people will experience them every now and again. Anyone can get a nosebleed, but they most often affect:

- children between two and 10 years of age
- elderly people
- pregnant women
- people who take blood thinning medication such as aspirin or anticoagulants, such as warfarin
- people with blood clotting disorders, such as haemophilia

Bleeding may also be heavier or last longer if you take anticoagulants, have a blood clotting disorder, or have high blood pressure (hypertension).

Are nosebleeds serious?

Nosebleeds aren't usually serious. However, frequent or heavy nosebleeds may indicate more serious health problems, such as high blood pressure or a blood clotting disorder, and should be checked.

Excessive bleeding over a prolonged period of time can also lead to further problems such as anaemia.

If your GP suspects a more serious problem is causing your nosebleeds, they may refer you to an ear, nose and throat (ENT) specialist for further tests.

Preventing nosebleeds

Things you can do to prevent nosebleeds include:

- avoid picking your nose and keep your fingernails short
- blow your nose as little as possible and only very gently
- keep your home humidified
- wear a head guard during activities in which your nose or head could get injured

- always follow the instructions that come with nasal decongestants – overusing these can cause nosebleeds

Talk to your doctor if you experience nosebleeds frequently and aren't able to prevent them. They may refer you to an ENT specialist for an assessment.

Management of Cuts

These guidelines can help you care for minor cuts and scrapes:

1. **Wash your hands.** This helps avoid infection.
2. **Stop the bleeding.** Minor cuts and scrapes usually stop bleeding on their own. If needed, apply gentle pressure with a clean bandage or cloth and elevate the wound until bleeding stops.
3. **Clean the wound.** Rinse the wound with water. Keeping the wound under running tap water will reduce the risk of infection. Wash around the wound with soap. But don't get soap in the wound. And don't use hydrogen peroxide or iodine, which can be irritating. Remove any dirt or debris with a tweezers cleaned with alcohol. See a doctor if you can't remove all debris.
4. **Apply an antibiotic or petroleum jelly.** Apply a thin layer of an antibiotic ointment or petroleum jelly to keep the surface moist and help prevent scarring. Certain ingredients in some ointments can cause a mild rash in some people. If a rash appears, stop using the ointment.
5. **Cover the wound.** Apply a bandage, rolled gauze or gauze held in place with paper tape. Covering the wound keeps it clean. If the injury is just a minor scrape or scratch, leave it uncovered.
6. **Change the dressing.** Do this at least once a day or whenever the bandage becomes wet or dirty.
7. **Get a tetanus shot.** Get a tetanus shot if you haven't had one in the past five years and the wound is deep or dirty.
8. **Watch for signs of infection.** See a doctor if you see signs of infection on the skin or near the wound, such as redness, increasing pain, drainage, warmth or swelling.

Management of Fractures

FRACTURE



- 1 Do not move the casualty.
- 2 Support and do not move the injured part.
- 3 Send for medical help.

FIRST AID AS EASY AS 1-2-3!

Figure 9: Fracture First Aid

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A fracture is a broken bone. It requires medical attention. If the broken bone is the result of major trauma or injury, call 911 or your local emergency number.

Also call for emergency help if:

- The person is unresponsive, isn't breathing or isn't moving. Begin CPR if there's no breathing or heartbeat.
- There is heavy bleeding.
- Even gentle pressure or movement causes pain.
- The limb or joint appears deformed.
- The bone has pierced the skin.
- The extremity of the injured arm or leg, such as a toe or finger, is numb or bluish at the tip.
- You suspect a bone is broken in the neck, head or back.

Don't move the person except if necessary to avoid further injury. Take these actions immediately while waiting for medical help:

- **Stop any bleeding.** Apply pressure to the wound with a sterile bandage, a clean cloth or a clean piece of clothing.
- **Immobilize the injured area.** Don't try to realign the bone or push a bone that's sticking out back in. If you've been trained in how to splint and professional help isn't readily available, apply a splint to the area above and below the fracture sites. Padding the splints can help reduce discomfort.

- **Apply ice packs to limit swelling and help relieve pain.** Don't apply ice directly to the skin. Wrap the ice in a towel, piece of cloth or some other material.
- **Treat for shock.** If the person feels faint or is breathing in short, rapid breaths, lay the person down with the head slightly lower than the trunk and, if possible, elevate the legs.

Management of Drowning

Call 999 if:

- Someone is drowning

1. Get Help

- Notify a lifeguard, if one is close. If not, ask someone to call 911.
- If you are alone, follow the steps below.

2. Move the Person

- Take the person out of the water.

3. Check for Breathing

- Place your ear next to the person's mouth and nose. Do you feel air on your cheek?
- Look to see if the person's chest is moving.

4. If the Person is Not Breathing, Check Pulse

- Check the person's pulse for 10 seconds.

5. If There is No Pulse, Start CPR

Carefully place person on back.

- For an adult or child, place the heel of one hand on the center of the chest at the nipple line. You can also push with one hand on top of the other. For an infant, place two fingers on the breastbone.
- For an adult or child, press down at least 2 inches. Make sure not to press on ribs. For an infant, press down about 1 and 1/2 inches. Make sure not to press on the end of the breastbone.
- Do chest compressions only, at the rate of 100-120 per minute or more. Let the chest rise completely between pushes.
- Check to see if the person has started breathing.

6. Repeat if Person Is Still Not Breathing

- If you've been trained in CPR, you can now open the airway by tilting the head back and lifting the chin.
- Pinch the nose of the victim closed. Take a normal breath, cover the victim's mouth with yours to create an airtight seal, and then give 2 one-second breaths as you watch for the chest to rise.
- Give 2 breaths followed by 30 chest compressions.

- Continue this cycle of 30 compressions and 2 breaths until the person starts breathing or emergency help arrives.

Management of Poisoning

Poisoning is injury or death due to swallowing, inhaling, touching or injecting various drugs, chemicals, venoms or gases. Many substances — such as drugs and carbon monoxide — are poisonous only in higher concentrations or dosages. And others — such as cleaners — are dangerous only if ingested. Children are particularly sensitive to even small amounts of certain drugs and chemicals.

How you treat someone who may have been poisoned depends on:

- The person's symptoms
- The person's age
- Whether you know the type and amount of the substance that caused poisoning

If you are concerned about possible poisoning, call Poison Help at 800-222-1222 in the United States or your regional poison control center. Poison control centers are excellent resources for poisoning information and, in many situations, may advise that in-home observation is all that's needed.

When to suspect poisoning

Poisoning signs and symptoms can mimic other conditions, such as seizure, alcohol intoxication, stroke and insulin reaction. Signs and symptoms of poisoning may include:

- Burns or redness around the mouth and lips
- Breath that smells like chemicals, such as gasoline or paint thinner
- Vomiting
- Difficulty breathing
- Drowsiness
- Confusion or other altered mental status

If you suspect poisoning, be alert for clues such as empty pill bottles or packages, scattered pills, and burns, stains and odors on the person or nearby objects. With a child, consider the possibility that he or she may have applied medicated patches or swallowed a button battery.

When to call for help

Call 911 or your local emergency number immediately if the person is:

- Drowsy or unconscious
- Having difficulty breathing or has stopped breathing
- Uncontrollably restless or agitated
- Having seizures
- Known to have taken medications, or any other substance, intentionally or accidentally overdosed (in these situations the poisoning typically involves larger amounts, often along with alcohol).

Call Poison Help at 800-222-1222 in the United States or your regional poison control center in the following situations:

- The person is stable and has no symptoms
- The person is going to be transported to the local emergency department

Be ready to describe the person's symptoms, age, weight, other medications he or she is taking, and any information you have about the poison. Try to determine the amount ingested and how long since the person was exposed to it. If possible, have on hand the pill bottle, medication package or other suspect container so you can refer to its label when speaking with the poison control center.

What to do while waiting for help

Take the following actions until help arrives:

- **Swallowed poison.** Remove anything remaining in the person's mouth. If the suspected poison is a household cleaner or other chemical, read the container's label and follow instructions for accidental poisoning.
- **Poison on the skin.** Remove any contaminated clothing using gloves. Rinse the skin for 15 to 20 minutes in a shower or with a hose.
- **Poison in the eye.** Gently flush the eye with cool or lukewarm water for 20 minutes or until help arrives.
- **Inhaled poison.** Get the person into fresh air as soon as possible.
- If the person vomits, turn his or her head to the side to prevent choking.
- Begin CPR if the person shows no signs of life, such as moving, breathing or coughing.
- Call Poison Help at 800-222-1222 in the United States or your regional poison control for additional instructions.
- Have somebody gather pill bottles, packages or containers with labels, and any other information about the poison to send along with the ambulance team.

Caution

- **Syrup of ipecac.** Don't give syrup of ipecac or do anything to induce vomiting. Expert groups, including the American Association of Poison Control Centers and the American Academy of Pediatrics, no longer endorse using ipecac in children or adults who have taken pills or other potentially poisonous substances. No good evidence proves its effectiveness, and it often can do more harm than good.

If you still have old bottles of syrup of ipecac in your home, throw them away.

- **Button batteries.** The small, flat batteries used in watches and other electronics — particularly the larger, nickel-sized ones — are especially dangerous to small children. A battery stuck in the esophagus can cause severe burns in as little as 2 hours.

If you suspect that a child has swallowed one of these batteries, immediately take him or her for an emergency X-ray to determine its location. If the battery is in the esophagus, it will have to be removed. If it has passed into the stomach, it's usually safe to allow it to pass on through the intestinal tract.

- **Medicated patches.** If you think a child got hold of medicated patches (adhesive products for transdermal drug delivery), carefully inspect the child's skin and remove any

that are attached. Also check the roof of the mouth, where they can get stuck if the child sucks on them.

Management of Snake Bites

Most snakes aren't dangerous to humans. Only about 15% worldwide and 20% in the United States are venomous. In North America, these include the rattlesnake, coral snake, water moccasin and copperhead. Their bites can cause severe injuries and sometimes death.

Most venomous snakes have eyes like slits and are known as pit vipers. Their heads are triangular and they have fangs. One exception is the coral snake, which has a rounded head and round pupils. Nonvenomous snakes typically have rounded heads, round pupils and no fangs.

If a venomous snake bites you, call 911 or your local emergency number immediately, especially if the bitten area changes color, begins to swell or is painful. Many emergency rooms stock antivenom drugs, which may help you.

If possible, take these steps while waiting for medical help:

- Move beyond the snake's striking distance.
- Remain still and calm to help slow the spread of venom.
- Remove jewelry and tight clothing before you start to swell.
- Position yourself, if possible, so that the bite is at or below the level of your heart.
- Clean the wound with soap and water. Cover it with a clean, dry dressing.

Caution

- Don't use a tourniquet or apply ice.
- Don't cut the wound or attempt to remove the venom.
- Don't drink caffeine or alcohol, which could speed your body's absorption of venom.
- Don't try to capture the snake. Try to remember its color and shape so that you can describe it, which will help in your treatment. If you have a smartphone with you and it won't delay your getting help, take a picture of the snake from a safe distance to help with identification.

Symptoms

Most snakebites occur on the extremities. Typical symptoms of the bite from a nonvenomous snake are pain and scratches at the site.

Usually, after a bite from a venomous snake, there is severe burning pain at the site within 15 to 30 minutes. This can progress to swelling and bruising at the wound and all the way up the arm or leg. Other signs and symptoms include nausea, labored breathing and a general sense of weakness, as well as an odd taste in the mouth.

Some snakes, such as coral snakes, have toxins that cause neurological symptoms, such as skin tingling, difficulty speaking and weakness.

Sometimes, a venomous snake can bite without injecting venom. The result of these "dry bites" is irritation at the site.

To watch a video follow the link below

<https://www.youtube.com/watch?v=zINn6wXwV08>

Assisting an Asthma Attack Patient

Diagnosis

For adults and children over 5 years old, lung (pulmonary) function tests are used to check how well the lungs are working. Poor lung function is a sign that your asthma isn't well controlled. In some cases, lung function tests are also used in asthma emergencies to help your doctor understand the severity of an asthma attack or how well treatment is working.

Lung function tests include:

- **Peak flow.** Your doctor may take a peak flow reading when you come in for a scheduled visit or for emergency treatment during an asthma attack. This test measures how quickly you can breathe out. You may also use a peak flow meter at home to monitor your lung function.
The results of this test are known as peak expiratory flow (PEF). A peak flow test is done by blowing into a mouthpiece as hard and as fast as you can with a single breath (expiration).
- **Spirometry.** During spirometry, you take deep breaths and forcefully exhale into a hose connected to a machine called a spirometer. A common spirometry measurement is forced expiratory volume, which measures how much air you can breathe out in one second.
The results of this test are known as forced expiratory volume (FEV). Spirometry can also measure how much air your lungs can hold and the rate at which you can inhale and exhale.
- **Nitric oxide measurement.** This exam measures the amount of nitric oxide gas you have in your breath when you exhale. High nitric oxide readings indicate inflammation of the bronchial tubes.
- **Pulse oximetry.** This test measures the amount of oxygen in your blood. It's measured through your fingernail and only takes seconds.

Management of Asthma Attack

If you and your doctor have worked out an asthma plan, follow its directions at the first sign of an asthma attack.

This generally means taking two to six puffs of a quick-acting (rescue) inhaler to get airway-expanding medication, such as albuterol (ProAir HFA, Proventil HFA, Ventolin HFA, others) and levalbuterol (Xopenex), deep into your lungs. Small children and those who have trouble with inhalers can use a nebulizer. After 20 minutes, you can repeat the treatment one time if necessary. If you continue to wheeze or feel breathless after treatment, visit your doctor or urgent care that day.

If you're having symptoms of a severe asthma attack, such as difficulty speaking because you're so short of breath, use your quick-acting (rescue) medication and get to a doctor's office or urgent care immediately.

Your doctor may recommend that you continue to use quick-acting medication every three to four hours for a day or two after the attack. You might also need to take oral corticosteroid medication for a short time.

Emergency Management of Asthma Attack

If you go to the emergency room for an asthma attack in progress, you'll need medications to get your asthma under immediate control. These can include:

- **Short-acting beta agonists, such as albuterol.** These are the same medications as those in your quick-acting (rescue) inhaler. You may need to use a machine called a nebulizer, which turns the medication into a mist that can be inhaled deep into your lungs.
- **Oral corticosteroids.** Taken in pill form, these medications help reduce lung inflammation and get your asthma symptoms under control. Corticosteroids can also be given intravenously, typically to patients who are vomiting or who are experiencing respiratory failure.
- **Ipratropium (Atrovent HFA).** Ipratropium is sometimes used as a bronchodilator to treat a severe asthma attack, especially if albuterol is not fully effective.
- **Intubation, mechanical ventilation and oxygen.** If your asthma attack is life-threatening, your doctor may put a breathing tube down your throat into your upper airway. Using a machine that pumps oxygen into your lungs will help you breathe while your doctor gives you medications to bring your asthma under control.

After your asthma symptoms improve, your doctor may want you to stay in the emergency room for a few hours or longer to make sure you don't have another asthma attack. When your doctor feels your asthma is sufficiently under control, you'll be able to go home. Your doctor will give you instructions on what to do if you have another attack.

If your asthma symptoms don't improve after emergency treatment, your doctor may admit you to the hospital and give you medications every hour or every few hours. If you're having severe asthma symptoms, you may need to breathe oxygen through a mask. In some cases, a severe, persistent asthma attack requires a stay in the intensive care unit (ICU).

Assisting an Epileptic Patient

Epilepsy is a common condition of the brain in which a person tends to have recurrent unprovoked seizures.

About 70% of people with epilepsy gain control of their seizures with medication. People who continue to have seizures are more vulnerable to the potential risks associated with seizures, especially when seizures occur without warning and impair awareness.

Epilepsy, like other long-term conditions such as asthma or diabetes, comes with certain risks. If left unchecked these can become very serious.

Seizure-related risks are higher when people have poorly controlled seizures. Good seizure control is the first step in reducing seizure-related risks.

Seizures can sometimes lead to injuries or falls, and they can occasionally be more serious – even contributing to, or causing, death. Different types of seizures carry different risks.

Your risk level depends on the type of seizures you have, and your lifestyle. For instance, poorly controlled tonic-clonic seizures pose the highest safety risk, and if you take part in activities such as mountain climbing, this risk is increased.

Life is never risk-free, but taking positive action to reduce your seizures, thinking about risks specific to you and discussing seizure management with your doctor are a start to reducing some of your seizure-related risks.

Epileptic seizure first aid

If you are with someone having a tonic-clonic seizure (where the body stiffens, followed by general muscle jerking), try to:

- Stay calm and remain with the person.
- If they have food or fluid in their mouth, roll them onto their side immediately.
- Keep them safe and protect them from injury.
- Place something soft under their head and loosen any tight clothing.
- Reassure the person until they recover.
- Time the seizure, if you can.
- Gently roll the person onto their side after the jerking stops.

Do not put anything into their mouth or restrain or move the person, unless they are in danger.

If a person having a seizure is in a wheelchair

If a person has a seizure when they are in a wheelchair, car seat or stroller:

- Leave the person seated with the seatbelt on (unless it is causing injury).
- Put the wheelchair brakes on.
- If it's a tilt wheelchair, tilt the seat and lock in position.

- Support their head until the seizure has ended.
- Lean the person slightly to one side to aid drainage of any fluid in the mouth.

After the seizure, if the person is having trouble breathing or they need to sleep, take them out of the chair and put them in the recovery position.

If their breathing difficulties continue, call an ambulance and closely monitor the person. Be prepared to perform **CPR** if they stop breathing.

Be aware that the **CPR technique for adults** is different to the **CPR technique for young children and infants**.

Seizures in water

If a seizure occurs in water:

1. Support the person's head so their face is out of the water.
2. Tilt their head back to ensure a clear airway.
3. If the person is in a pool, remove them from the water when the jerking stops. Note: In the rare circumstance the jerking does not stop, seek help from others if possible, and remove the person from the water at the shallowest end of the pool.
4. If the person is in the surf, remove them from the water immediately.
5. Flotation devices may be useful when removing someone from water.
6. Seek help if possible.

Once out of the water:

- Call triple zero (000) for an ambulance immediately. (Do this even if the person is breathing, as they may have inhaled water)
- Place the person on their side.
- Check to see if they are breathing.
- If they are not breathing, or they are not breathing normally, reposition the person onto their back and begin the appropriate form of CPR: **CPR for adults** is different to the **CPR for young children and infants**.

When to call an ambulance for an epileptic seizure

Call for an ambulance if:

- the seizure lasts for **5** or more minutes or longer than what is normal for the person
- a second seizure quickly follows
- the person is not responding for more than 5 minutes after the seizure ends
- the person has breathing difficulties after the jerking stops

- it is the person's first known seizure
- the seizure occurs in water
- the person is injured
- you are in doubt.

Assisting a Diabetic Patient

Diabetes can become serious in the short term if blood sugar levels become either too high or too low. The following information details what to do in an emergency.

This covers low blood sugar (hypoglycemia), very high blood sugar (diabetic ketoacidosis) and what to do if you are left without your diabetes medication and/or supplies.

What counts as a diabetic emergency?

It can be a difficult area sometimes to know what counts as a genuine emergency.

News reports in recent years have highlighted that a significant number of '999' ambulance call-outs have not been necessary – for example to treat mild hypoglycemia which, in some cases, has been successfully treated before the ambulance has arrived.

This isn't to say that conditions, such as hypoglycemia, are not dangerous but that it's important to know when a situation really is an emergency so that an ambulance is not unnecessarily called.

When should I call an ambulance?

An ambulance will be needed if someone has either very high or very low blood sugar levels that presents an immediate danger and neither they nor anyone around is confidently able to treat them.

Ketoacidosis and Hyperosmolar Hyperglycemic Nonketotic Syndrome are both life threatening conditions. Hypoglycemia can also be life threatening in some cases.

Someone with diabetes that is unconscious is one of the situations in which you should call for an ambulance.

Severe hypoglycemia

Hypoglycemia can become dangerous if it is not treated quickly, particularly if it is a result of an insulin overdose.

Severe hypoglycemia is generally recognised as hypoglycemia involving:

- Convulsions (fitting)
- Unconsciousness

Hypoglycemia can often be treated at home but if someone is unable to take sugar or does not show signs of improvement after taking sugar, other options will be needed such as a glucagon injection or an ambulance.

If you have glucagon, check it is in date, and follow the instructions in the glucagon kit carefully. If you are unsure about using the kit, and nobody is present who is, call for an ambulance.

Treating non-severe hypoglycaemia

- To prevent hypoglycemia developing into a more serious situation, make sure you have hypo treatments, such as the HypoWallet, to hand.

Preventing emergencies

In most cases, non-severe hypoglycemia can be treated by taking glucose followed by slower acting carbohydrate such as bread or fruit.

To prevent hypoglycemia developing into a more serious situation, make sure you have hypo treatments available at all times. Having a HypoWallet is a useful way to ensure you're well prepared for hypos.

Diabetic ketoacidosis (DKA)

Diabetic ketoacidosis is a serious condition which develops if blood glucose levels become very high.

The symptoms include vomiting and a laboured form of breathing known as kussmaul breathing, which may be identified by gasping breaths. If someone is going through DKA, it is often possible to detect ketones on the breath, which can be similar to a fruity smell or similar to nail polish.

If diabetic ketoacidosis is suspected and the person is either incoherent or unable to help him or herself, call 999.

Hyperosmolar Hyperglycemic Nonketotic Syndrome (HHNS)

HHNS is a short term complication that can occur in people with non-insulin dependent diabetes, such as type 2 diabetes, if blood glucose levels rise to a very high level of 30 mmol/l or above.

If someone is showing the signs of HHNS it is important they receive emergency treatment.

Whilst the ambulance is coming

If the person is unconscious or unable to sit or stand, put them into the recovery position. Try to stay with the person in case their situation should get worse.

Diabetes emergency ID

If an emergency situation is developing, it's important the people around you know that you have diabetes to help them to be able to help you.

Wearing emergency identification will help people, including paramedics, to recognise that you have diabetes should you need emergency care.

- Alert ID diabetes emergency identification

Running out of diabetes medication or supplies

You should ensure you have sufficient supplies, including spares, wherever possible but there can be times when you may need emergency medication or supplies. If you need medication and supplies right away you will need to get an emergency prescription.

Effective communication on casualty's condition

Thinking about how best to communicate with casualties requires an understanding of how people behave in emergencies, what they need to hear, and how they want to hear it. Answers to these questions can come from those who have experienced being casualties, but this comes with significant challenges. A useful alternative is to collect feedback from people who have played casualties in live exercises.

Two key positive factors in Casualty Communication

- Reassurance – the ability to reassure and calm the individual down
- Authoritative directness – the ability to give simple, clear and directive instructions

To assist frontline responders in remembering key methods to communicate we recommend bearing in mind the following **'FEAR'** mnemonic:

F – communicate clear facts about what has happened, when responders can get to casualties, what is going on, etc. And if you aren't clear make it clear you aren't clear!

E – establish a line of communication as early as humanly possible.

A – where possible keep casualties active in their own recovery and mobilisation. Research indicates a major factor in saving lives is the active participation of others in the incident in providing assistance.

R – updates and communication must be regular. Even if there is nothing to update on, just ensuring that casualties are told 'nothing has changed but we are still doing X Y and Z' creates an ongoing and important dialogue. Don't leave casualties for unspecified amounts of time and always indicate roughly when you will update them.

Handing over casualty to medical personnel

Once you have connected with the emergency service's operator and requested an ambulance, hold the line to tell them details such as:

- Location of casualty—providing the exact address is best, but if you aren't sure of this, give some landmarks or nearest cross streets, and some directions
- Number of casualties
- Nature and extent of illness or accident—include the physical condition of the casualty, and any relevant signs and symptoms (see below for more details)
- The number of the phone you are using
- Your name.

Then hold the line to answer any further questions and provide any other relevant details, such as damaged power lines.

The casualty's condition

When providing details of the casualty's condition, include the following

- Colour—What colour is the casualty's skin?
- Conscious state—Is the casualty conscious?
- Breathing—Is the casualty breathing? If not, have rescue breaths and CPR commenced?
- Bleeding—Is the bleeding controlled? Is it bright red spurting blood or dark red flowing blood?
- Pupils—Are the pupils of equal size and reacting to light?
- Shock—Are there signs of shock (pale, sweaty, nauseous, cold)?
- Movement—Does the casualty have coordinated movement?
- Can the casualty feel and move limbs, if conscious?
- Anything abnormal (such as bruising, a bone sticking out or swelling).

Handing over to Emergency Services personnel

On the arrival of the ambulance:

- Continue first aid management and observation of the casualty until the ambulance officer is ready to assume care
 - provide as much information as possible, this may include:
 - the nature of the accident to the time you arrived on the scene
 - the types of first aid management provided
 - the duration of any Basic Life Support
- Provide any other information that is asked for, including your contact details
- Stay and assist the ambulance officer if requested to do so.

Monitoring Casualty's condition

It is important that the whole process of responding to the casualty occurs quickly so that assessment and management of the situation may be carried out effectively.

How do you know that somebody is in need of first aid in the first place? What are the signs that make you suddenly stop what you are doing and become more alert to an incident or situation? Your answer may include using your senses:

What do you hear? Perhaps you heard shouting, screaming, a call for help, a crash, or moaning.

What do you see? Did you see a person on the ground, strange behaviour, an upturned chair, a group of people, smoke or flames?

What do you smell? Did you smell gas, petrol, fumes or burning?

The following are some guidelines for responding to the casualty's condition:

- Do not approach the casualty if it is unsafe to do so.
- ALWAYS call for help.
- If the casualty is unconscious, determine airway and breathing
- If the person is conscious but can't move or feel their arms or legs, DON'T move them. They may have a spinal injury.
- Manage illness and injuries with the guidance of a qualified first aider and stay with the casualty.

If you need to phone for an ambulance and give details of the situation and workplace location, make sure you know exactly where you are! You should know the following:

- Address
- Nearest cross street
- The building level you are on
- The best way for the ambulance to gain access to the casualty

Effective first aid principles

What actually is first aid? First aid is the initial care of the sick or injured. First aid management is a set of established procedures and aims to:

- Help promote recovery
- Prevent the injury or sickness from becoming worse
- Seek medical help
- Promote a safe environment
- Provide comfort
- Protect the unconscious casualty
- Preserve life.

It is important to follow these basic first aid steps:

- Do not panic
- Assess the situation quickly
- Make sure the area is safe to approach
- Identify the injury or sickness
- Call for help
- Stay with the casualty
- Manage the casualty
- Monitor the casualty; their situation could change at any time

4.2.1.4 Learning Activities

4.2.1.4.1 Practical Activities

1. You are in the skills laboratory and your instructor provides you with a mannequin.

Demonstrate how to provide first aid to an epileptic patient using the mannequin?

4.2.1.4.2 Self-Assessment

You are provided with the following questions for self -assessment, attempt them and check your responses

1. You have encountered an unconscious patient who needs cardio-pulmonary resuscitation. What are the steps of cardio-pulmonary resuscitation?
2. You are the only medic around in your village. Your neighbor calls you to help an adult who has experienced severe choking. What is the procedure of performing back blows and abdominal thrusts?
3. What is the management of burns and scalds?
4. You are the first to reach to the scene of a patient involved in trauma. Primary survey in the management of trauma involves?
5. One of your classmates begins nose bleeding. Demonstrate how you will help him to stop the nose bleeding?
6. Cuts and scrapes are very common in day to day life. How can you help a patient with a minor cut or scrapes?
7. One of your friends drowns in a swimming escapade. As a health worker, how can you help a drowning patient?
8. You are in the emergency room where you have a patient with asthma attack. What is the emergency management of asthma attack?

4.2.1.5 Tools, Equipment, Supplies and Materials

- Casualty department
- Functional skills laboratory
- First aid kit
- Emergency room
- Emergency room standard manuals
- Emergency trolley
- First Aid Manual.
- Tweezers
- Alcohol Swabs
- Antibiotic Ointment
- Bandages
- Gauze Pads
- Medical Tape
- Elastic Bandages

- Pain relievers
- Instant cold pack
- Waste disposal bins
- Personal protective equipment
- Health workers
- Presentation charts
- Flip charts

4.2.1.6 References

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2.2.1.7 MODEL ANSWERS

Self-Assessment

1. You have encountered an unconscious patient who needs cardio-pulmonary resuscitation. What are the steps of cardio-pulmonary resuscitation?

- Step 1 Call 999 or the nearest ambulance services
- Step 2 Place the person on their back and open their airway
- Step 3 Check for breathing-Place your ear next the person's mouth and listen for no more than 10 seconds. If you do not hear breathing, or you only hear occasional gasps, begin CPR.
- Step 4 Perform 30 chest compressions-Place one of your hands on top of the other and clasp them together. With the heel of the hands and straight elbows, push hard and fast in the center of the chest, slightly below the nipples.
- Push at least 2 inches deep. Compress their chest at a rate of least 100 times per minute. Let the chest rise fully between compressions.
- Step 5 Perform two rescue breaths-Making sure their mouth is clear, tilt their head back slightly and lift their chin. Pinch their nose shut, place your mouth fully over theirs, and blow to make their chest rise.
- Step 6 Repeat-Repeat the cycle of 30 chest compressions and two rescue breaths until the person starts breathing or help arrives.

2. You are the only medic around in your village. Your neighbor calls you to help an adult who has experienced severe choking. What is the procedure of performing back blows and abdominal thrusts?

Back blows Procedure

- Stand behind them and slightly to one side. Support their chest with 1 hand. Lean them forward so the object blocking their airway will come out of their mouth, rather than moving further down.
- Give up to 5 sharp blows between their shoulder blades with the heel of your hand. The heel is between the palm of your hand and your wrist.
- Check if the blockage has cleared.
- If not, give up to 5 abdominal thrusts.

Abdominal Thrust Procedure

- Stand behind the person who's choking.
- Place your arms around their waist and bend them forward.
- Clench 1 fist and place it right above their belly button.
- Put the other hand on top of your fist and pull sharply inwards and upwards.
- Repeat this movement up to 5 times.

3. What is the management of burns and scalds?

- immediately get the person away from the heat source to stop the burning

- cool the burn with cool or lukewarm running water for 20 minutes – don't use ice, iced water, or any creams or greasy substances such as butter
 - remove any clothing or jewellery that's near the burnt area of skin, including babies' nappies - but don't move anything that's stuck to the skin
 - make sure the person keeps warm – by using a blanket, for example, but take care not to rub it against the burnt area
 - cover the burn by placing a layer of cling film over it – a clean plastic bag could also be used for burns on your hand
 - use painkillers such as paracetamol or ibuprofen to treat any pain
 - if the face or eyes are burnt, sit up as much as possible, rather than lying down - this helps to reduce swelling
4. **You are the first to reach to the scene of a patient involved in trauma. Primary survey in the management of trauma involves?**

- **Airway**-- open the patient's airway using a jaw-thrust maneuver with manual, in-line stabilization of the neck. If you find food, blood, vomitus, or other debris, suction the airway quickly to prevent aspiration. To better remove secretions, you may need to carefully logroll the patient to his side. Manually stabilize his neck and spine as you do so.
 - **Breathing**-- Note respiratory rate and depth, chest expansion, and accessory muscle use and auscultate breath sounds bilaterally. Also palpate for crepitus or subcutaneous air in the neck and chest, which can indicate a pneumothorax or airway injury.
 - **Circulation**--Once you've assessed and supported your patient's breathing, attend to his circulatory status. Assess for the presence and quality of peripheral pulses to quickly estimate blood pressure
 - **Disability**--To evaluate disability, you'll evaluate the patient's LOC, pupil response, and gross sensorimotor function. To document his baseline LOC, quickly assess and record an initial GCS score.
 - **Exposure**--The final component of the primary survey is *exposure*. Remove the patient's clothing completely so you can inspect his entire body for injuries. Use good judgment when removing clothing; trying to remove a shirt by pulling or manipulating it may worsen the injury or pain. Cutting clothing away with trauma shears is usually best.
5. **One of your classmates begins nose bleeding. Demonstrate how you will help him to stop the nose bleeding?**
- sit down and firmly pinch the soft part of your nose, just above your nostrils, for at least 10-15 minutes
 - lean forward and breathe through your mouth – this will drain blood into your nose instead of down the back of your throat
 - place an ice pack or bag of frozen vegetables covered by a towel on the bridge of your nose

- stay upright, rather than lying down, as this reduces the blood pressure in the blood vessels of your nose and will discourage further bleeding
- If the bleeding eventually stops, you won't usually need to seek medical advice. However, in some cases you may need further treatment from your doctor or in hospital

6. Cuts and scrapes are very common in day to day life. How can you help a patient with a minor cut or scrapes?

- Wash your hands. This helps avoid infection.
- Stop the bleeding. Minor cuts and scrapes usually stop bleeding on their own. If needed, apply gentle pressure with a clean bandage or cloth and elevate the wound until bleeding stops.
- Clean the wound. Rinse the wound with water. Keeping the wound under running tap water will reduce the risk of infection. Wash around the wound with soap. But don't get soap in the wound. And don't use hydrogen peroxide or iodine, which can be irritating. Remove any dirt or debris with a tweezers cleaned with alcohol. See a doctor if you can't remove all debris.
- Apply an antibiotic or petroleum jelly. Apply a thin layer of an antibiotic ointment or petroleum jelly to keep the surface moist and help prevent scarring. Certain ingredients in some ointments can cause a mild rash in some people. If a rash appears, stop using the ointment.
- Cover the wound. Apply a bandage, rolled gauze or gauze held in place with paper tape. Covering the wound keeps it clean. If the injury is just a minor scrape or scratch, leave it uncovered.
- Change the dressing. Do this at least once a day or whenever the bandage becomes wet or dirty.
- Get a tetanus shot. Get a tetanus shot if you haven't had one in the past five years and the wound is deep or dirty.
- Watch for signs of infection. See a doctor if you see signs of infection on the skin or near the wound, such as redness, increasing pain, drainage, warmth or swelling.

7. One of your friends drowns in a swimming escapade. As a health worker, how can you help a drowning patient?

- Get Help--Notify a lifeguard, if one is close. If not, ask someone to call 999.
- Move the Person--Take the person out of the water.
- Check for Breathing--Place your ear next to the person's mouth and nose. Do you feel air on your cheek?
- If the Person is Not Breathing, Check Pulse--Check the person's pulse for 10 seconds.
- If There is No Pulse, Start cardio-pulmonary resuscitation
- Repeat cardio-pulmonary resuscitation if Person Is Still Not Breathing

8. You are in the emergency room where you have a patient with asthma attack. What is the emergency management of asthma attack?

- Short-acting beta agonists, such as albuterol--These are the same medications as those in your quick-acting (rescue) inhaler. You may need to use a machine called a nebulizer, which turns the medication into a mist that can be inhaled deep into your lungs.
- Oral corticosteroids. Taken in pill form, these medications help reduce lung inflammation and get your asthma symptoms under control. Corticosteroids can also be given intravenously, typically to patients who are vomiting or who are experiencing respiratory failure.
- Ipratropium (Atrovent HFA). Ipratropium is sometimes used as a bronchodilator to treat a severe asthma attack, especially if albuterol is not fully effective.
- Intubation, mechanical ventilation and oxygen. If your asthma attack is life-threatening, your doctor may put a breathing tube down your throat into your upper airway. Using a machine that pumps oxygen into your lungs will help you breathe while your doctor gives you medications to bring your asthma under control.

4.2.1 Learning Outcome 3:

Evaluate first aid intervention

4.2.1.1 Introduction to the learning outcome

This learning outcome specifies the content of competencies required to evaluate first aid intervention. It includes evaluating response of the casualty to the intervention and evaluating first aider's response to the situation. Effective evaluation of the casualty helps in proper response of the emergency situation.

4.2.1.2 Performance Standard

7. Evaluated the response of the casualty to the intervention based on the initial assessment
8. Evaluated the response of the first aider to the situation based on principles and procedures of first aid

4.2.1.3 Information Sheet

Definition of Terms

Casualty- the department of a hospital providing immediate treatment for emergency cases

First Aid- is the first and immediate assistance given to any person suffering from either a minor or serious illness or injury, with care provided to preserve life, prevent the condition from worsening, or to promote recovery

Emergency services- are the public organizations whose job is to take quick action to deal with emergencies when they occur, especially the fire brigade, the police, and the ambulance service.

Injury- also known as physical trauma is damage to the body caused by external force. This may be caused by accidents, falls, hits, weapons, and other causes.

Evaluating response of the casualty to the intervention

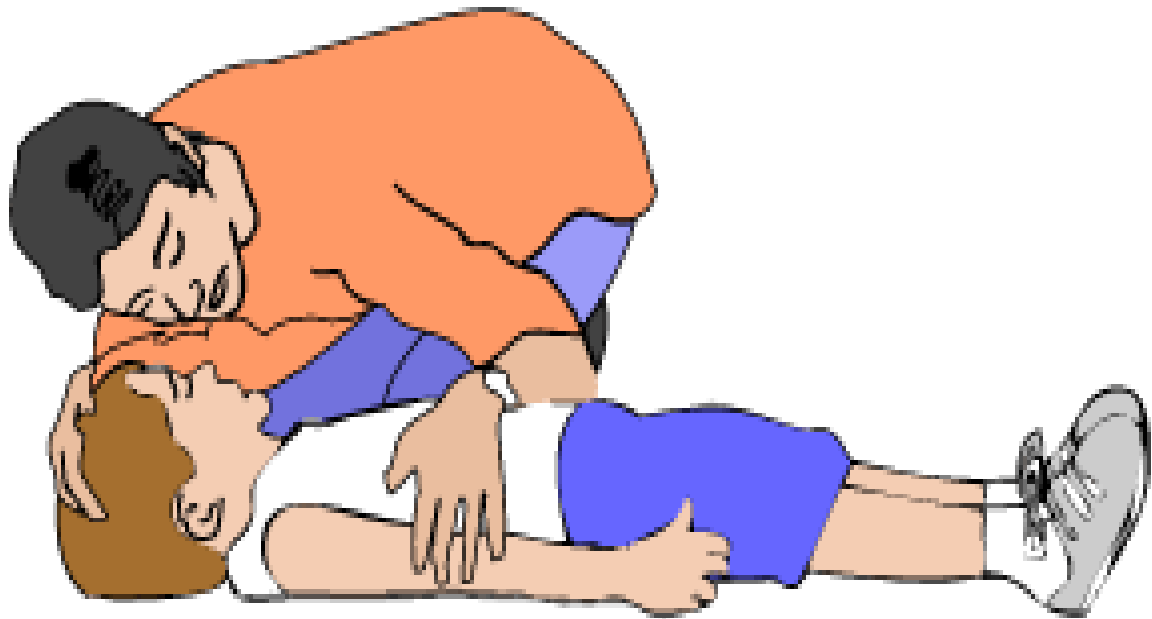


Figure 10: Apply First Aid, Assess Casualty

https://sielearning.tafensw.edu.au/MCS/FirstAid_Ultimo/firstaid/lo/5251/graphics/2006_157_f01.gif

An emergency plan is a set of arrangements for responding to, and recovering from emergencies, and it is about protecting life, property and the environment. The development of an emergency plan should take into account existing plans at other administrative levels, plans that operate at the same level, as well as any plans developed for specific hazards. The prerequisites for planning are: a recognition that risks and vulnerability exist, and that emergencies can occur; an awareness by the community, government, and decision-makers of the need to plan and of the benefits of planning; implementation of the plan is guaranteed by appropriate legislation; and, designation of an organization responsible for coordinating both planning and emergency response and recovery in the event of an emergency.

The planning process can be applied to any community, organization or activity.

It includes:

- **Project definition:** determines the aim, objectives and scope of an emergency plan, and decides the tasks and the resources required to performing these tasks.
- Formation of a representative **planning group:** to gather the required information and to gain the commitment of key people and organizations, both of which will contribute to the successful implementation of the plan.
- **Potential problem analysis:** through breaking the problem into its components to examine risks, their causes, possible preventive strategies, response and recovery strategies, and trigger events for these strategies.
- **Resource analysis:** to identify the required resources for the response and recovery strategies, resources available, discrepancy between requirement and availability, and responsibility.

- Designation of **roles and responsibilities** to individuals and organizations.
- **Management structure** concerning the command of individual organizations and control across organizations.
- Developing **strategies and systems** for specific response and recovery.
- **Documentation**: The written emergency plan will consist of the outputs of each step of the process.

A four-pillar strategy

The WHO's Health Action in Crises (HAC) has recently elaborated a five-year strategy to build the capacity of the health sector and communities in emergency preparedness and risk reduction. This Strategy consists of four pillars:

- Assessing and monitoring baseline information on the status of health emergency preparedness and risk management at regional and country levels
- Institutionalizing emergency preparedness and risk management in ministries of health and establishing an effective all-hazard/ whole-health programme for this purpose
- Encouraging and supporting community-based emergency preparedness and risk management
- Improving knowledge and skills in health emergency preparedness and response, and risk management.

This strategy aims to substantially increase the preparedness of the communities and the health sector to manage health consequences of mass casualty incidents and other emergencies and crises, and provides the conceptual underpinning to these Guidelines.

Guiding Principle

A number of guiding principles should be observed within all plans, protocols and operational procedures. These principles are designed to assist planners, policymakers, and implementing agencies as they develop plans for their own specific situations.

Clear lines of responsibility

Plans must clearly define the roles, responsibilities and expected activities of all those dealing with the incident. They must allow response to be scaled up from local, to provincial/state and to national levels in a seamless manner, with no confusion as to who is in charge at each phase of the response.

All-hazard applicability

While risk assessment will identify those hazards most likely to result in a mass casualty incident in a given community, that community must prepare its response in such a way that the arrangements will be effective no matter what the event, be it a hurricane, an airplane crash, or a pandemic.

Scalability

Preparedness must address different scales of incident and surge in demand for health care. While some activities (triage, transportation, treatment) are common to managing all mass casualty incidents, additional measures such as evacuation of large populations may be

necessary. For example, the scale of the 2004 Asian tsunami was beyond the thinking of planners. Plans must allow for “scale-up” in such events.

Whole-of-health

In addition to death and injury, other health considerations must be planned for. Planning strategies must also take into account the basics of environmental health (i.e., water, sanitation, housing); chronic diseases (including mental health); maternal and child health; communicable diseases; nutrition; and health care delivery services (including health infrastructure).

Knowledge-based

Almost every imaginable form of mass casualty incident has already occurred before, and planners therefore have access to a great – and growing – body of knowledge. Useful sources of information may include: official reports from other countries (particularly those with similar conditions); scientific and epidemiological data; and documentation from WHO and other organizations

Multisectoral

The success or failure of responses to mass casualty incidents depends on the cooperation of many sectors – communications, transportation, law and order, security, water and sanitation, social services, and other non-health sectors – that may not coordinate their “normal” daily activities. Coordination should be institutionalized not only at the level of national ministries, but – ultimately most important – at the local level of communities.

National policies which enable local solutions

The fundamental concerns of a mass casualty management system are public safety and the building of safer communities (“building” in this case means not only physical construction but strengthening all of the elements – human, organizational and infrastructural – that make up a community). While plans must be in place to mobilize provincial/state and national resources if required, the national policy and strategy emphasis must be clearly directed at enabling local authorities to prepare for, respond to and recover from a mass casualty incident.

No matter what forms that government and public administration take, national emergency management systems should include:

- Identification of lines of authority, from the national to the local level
- Financial arrangements for funding emergency work
- Arrangements to ensure that government and community activities are maintained (for example, creation of parallel or “hardened” communications systems to take over if normal voice or data transfer systems are affected)
- National stockpiling of appropriate resources (including provincial or state and local prepositioning of stockpiles)
- Database of national experts for advice on specific problems
- Protocols and formal arrangements for coordinated efforts with other countries, or between provincial/state governments within the country.

The Ministry of Health: a leading partner

Whatever the formal institutional arrangements, the Ministry of Health must be heavily involved in all national emergency management potentially dealing with mass casualties, because of both its expertise and its normative role in the setting of health-related standards and rules and procedures. For instance even if the Ministry of Health does not operate ambulances – in some countries this may be the task of the Red Cross/Red Crescent, Civil Defense authorities, or even the private sector – the Ministry of Health is nevertheless likely to set the standards regarding how ambulances are to be equipped, the training of crews, the numbers of ambulances per thousand population, etc. Because of its expertise, the Ministry of Health is usually well placed to help health care facilities and main health disciplines to develop their emergency plans, create training programme, prepare drills and programme, prepare drills and, prepare drills and exercises, and so on. Finally, the Ministry is usually responsible for promoting standards and overseeing accreditation, for example in validating the plans developed by health care facilities and their local partners.

A dedicated unit within the Ministry of Health

In order for the Ministry of Health to meet its potential as a partner, it must organize itself in a way which maximizes its comparative advantages and brings together the people best suited to the task at hand. A necessary first step will be to form permanent structures with institutional responsibility for all Ministry of Health activities related to planning for mass casualty events. A recommended configuration is a dedicated emergency committee of senior staff, chaired by the Minister of Health, and a dedicated emergency preparedness and response unit whose head serves as secretary of the Committee. The Committee and the Unit should adopt a multidisciplinary, all-hazard, whole-of-health approach, and therefore will consist of experts in all relevant fields.

Formal and informal relationships with other partners

In most countries, when a mass casualty incident occurs that requires a national response, some form of national emergency plan exists which is invoked according to established procedures. Within such arrangements, the immediate role of the Ministry of Health will be to act as a focal point for liaison, coordination and communication between the various health system components (e.g., public, private, military, and NGOs) as well as with other ministries, the office of the head of state and, when necessary, international agencies. This should be done based on a health sector plan (see following chapter) that takes into consideration the need for interdisciplinary coordination with health partners. Because of the potential for confusion arising over issues such as legal authority, jurisdiction, customs' importation procedures, and so on, the Ministry should sign Memoranda of Understanding (or equivalent agreements) with other partners to formalize the necessary arrangements. These agreements should be written with relevant national legislations in mind (see: A solid legislative basis, below).

Equally important, Ministry of Health staff should establish working relationships with staff in other organizations so that, during a crisis, the “players” (key individuals) are known to each other. This can be most effectively done by joint planning and exercises at all levels (discussed below), as well as regular meetings and periodic evaluations or reviews.

International cooperation

Although another part of government such as the Ministry of Foreign Affairs may have formal responsibility for requesting or accepting assistance from outside the country, the

Ministry of Health will have an essential role of facilitating, and coordinating international health assistance if it is required. This applies when assistance is offered or received, and will normally be coordinated with other ministries, the national Emergency Management Agency, and the office of the head of state. Once again, Memoranda of Understanding or other agreements for mutual aid, support and cooperation should be signed, particularly with neighboring countries, international agencies (including WHO) and international NGOs. It cannot be over-emphasized how important it is for these arrangements to have been prepared at a detailed level, in order for responses to work efficiently.

A solid legislative basis

The Ministry of Health should ensure (in concert with other ministries) that legislations exist which provides adequate authority for mass casualty management within the overall emergency preparedness and response plan. This goes well beyond providing special emergency powers when circumstances require, although that is of course an essential component. Responses from all levels of government (local, provincial/state, and national) must be defined and enabled in the appropriate laws, which will typically have titles like Emergency Management Act, Health Act, Community Services Act, and Quarantine Act. At a minimum, legislations should define the roles of Ministers and other senior officials (e.g., Chief Health Officer, Health/Medical Emergency Coordinator) and provide legal authority for national emergency councils, committees and advisory groups. It must enable the essential Command, Control and Coordination models which form the basis for all emergency management systems. If existing legislations or legislative gaps (i.e., lack of legally defined authority) act as barriers to effective mass casualty management, the Ministry of Health should help to draft the necessary amendments, according to the established law-making arrangements in the country.

Establishing the Baseline

Improving existing systems first requires establishing a “baseline” – an assessment of current system’s capacity, against which planned changes can eventually be measured. To do so, the Ministry of Health should perform a comprehensive analysis of relevant health care resources available in the country, or direct lower levels of government, such as provinces/states, to do so if that is where operational responsibility for health is located. This analysis may entail a comprehensive review of the health care system, or be a more tightly focused look at those components that will be particularly called on in the event of a mass casualty incident.

Whatever the focus, the baseline analysis should assess and map both the quantity and quality of available health care facilities, personnel and equipment.

Based on this analysis, a detailed risk assessment (see below), and consultation about the standards that risk assessment (see below), and consultation about the standards that must be met in emergency responses, a “gap analysis” can be carried out to (a) decide how to improve, a “gap analysis” can be carried out to (a) decide how to improve these resources to the required standards and (b) calculate the additional resources required to respond to a mass casualty incident should one occur. It will provide the Ministry of Health with the information it needs to set priorities and coordinate the redistribution of resources for optimum preparedness.

A variety of tools are available to assist with this process, and for assessing specific areas of health care.

For example, an analysis of the gap between existing trauma care resources and the desired standard can be performed using the WHO Guidelines for Essential Trauma Care and Pre-hospital Trauma Care

Systems (see Sources of Information). These publications set affordable minimum standards for trauma care capabilities, including human resources, physical resources and administrative mechanisms.

Hazard analysis and risk assessment

HAZARD RISK ASSESSMENT MATRIX

Frequency of Occurrence	Hazard Categories			
	1 Catastrophic	2 Critical	3 Serious	4 Minor
(A) Frequent	1A	2A	3A	4A
(B) Probable	1B	2B	3B	4B
(C) Occasional	1C	2C	3C	4C
(D) Remote	1D	2D	3D	4D
(E) Improbable	1E	2E	3E	4E



Unacceptable



High



Medium



Low

Table 2: Hazard Analysis and Risk Assessment Template

<https://i.pinimg.com/originals/fd/6a/78/fd6a78e9fb92272065658899aad006c6.jpg>

A second key initiative should be to provide a detailed understanding of the hazards and risks specific to the country. The WHO's Community emergency preparedness: a manual for managers and policy-makers provides a detailed discussion of this process. While it may be carried out or directed by the Ministry of Health at national level, hazard analysis and risk assessment should also be carried out on a provincial/state basis, particularly if health expertise and resources are concentrated at those levels. Emphasis should be placed on gathering retrospective data on previous incidents.

This should not be limited to large-scale disasters but to those of smaller scale such as major traffic or industrial accidents, mudslides or floods that did not require national intervention, fires and building collapses, and so on. It should also include a forward-looking component assessing risk in the future from elements such as unsafe urban development, or changes in weather patterns.

Coupled with the baseline assessment, the information provided by this hazard analysis and risk assessment will provide a solid basis for the subsequent process of planning, and for the allocation of resources. Having in hand the information collected in the baseline analysis, and in the hazard analysis and risk assessment, the Ministry of Health will be ready to help improve national preparedness for mass casualty incidents by implementing or sponsoring (i.e., helping other partners to implement) a number of initiatives. Some of these will be internal to the Ministry, while others will be carried out in cooperation with partner Ministries or agencies.

Monitoring, Surveillance and Early Warning

Ongoing collection and analysis of information is a vital part of preparedness. At the planning stage, the Ministry of Health should assess what types of information will be needed to support the smooth functioning of the National Mass Casualty Management Plan. Particular attention should be paid to the functions of monitoring, surveillance and early warning.

Monitoring system capacity

Unless some other part of government already has this responsibility, a unit within the Ministry (likely the one which carried out the baseline analysis) should be designated with responsibility to gather and maintain such information on an ongoing basis. This must include the most recent data on system capacity and resources, as well as documentation of lessons observed from previous incidents. As well as collecting such information, the unit should take responsibility for processing and disseminating such information in ways that are useful to policy makers and decision makers – both as part of preparation and in the event of a mass casualty incident.

The unit should make efforts to use information technology to best advantage. For example, Geospatial Information Systems (GIS) are increasingly affordable, and can be flexible and cost-effective means of both collecting and providing data vital to tasks such as establishment of temporary camps, quantifying deaths, and managing logistics. Similarly, web-based solutions (i.e., use of the internet) may be useful for involving and keeping stakeholders informed through dedicated websites, online courses and electronic discussion groups.

Surveillance, epidemiology and data collection

As noted earlier, mass casualty incident management does not solely involve trauma cases, although injuries do constitute an important part of the caseload in many incidents. With the increasing risk of pandemics and biological and chemical incidents, the Ministry of Health should ensure that a surveillance system is in place, which will undertake tasks such as pattern recognition in order to identify risks early and initiate protective measures as needed. However, it is important that surveillance systems routinely collect data on injuries since, as mentioned previously, most morbidity and mortality in mass casualty incidents may in fact come from the cumulative effects of smaller-scale events.

The operational requirements for the surveillance system should be informed by the hazard analysis and risk assessment exercise carried out earlier, which will have studied the epidemiology of the most prevalent types of incidents in the country. The Ministry of Health should continuously update this system to take into account new and emerging threats. As well as supporting preparedness, such a system will assist the efficient allocation and application of resources as an incident develops.

Early warning systems

As should be specified within the National Mass Casualty Management Plan (in coordination with national emergency management systems), that the surveillance function should include the provision of early warnings when potential incidents – including natural events such as floods and volcanic eruption – or other worrying trends are identified. This will require the creation of protocols for triggering warnings and the chain of command that needs to be informed.

National early warning systems should have agreements and working arrangements with their counterparts in neighboring countries, with regional structures (i.e., those grouping countries in cooperative arrangements), and with international organizations like WHO. This will permit efficient cooperation between countries should the scale or location of a given mass casualty incident warrant it.

As much as possible, this cooperation should have a practical orientation, with an emphasis on activities such as joint drills and exercises, regional training and technical networks. WHO Regional offices can facilitate such activities, offer training in relevant subjects, and obtain examples of plans or protocols being used in the region or internationally.

Financial and material resources

As well as creating a National Mass Casualty Management Plan, the Ministry of Health must ensure that it has done a detailed costing of the Plan, as well as estimating costs of responding to the most probable mass casualty incidents. This will include the resources used at local and provincial/state levels, as well as those for which the national level retains operational responsibility.

If the Ministry's current budget does not cover the necessary costs needed to conduct and support the activities specified in the Plan, the Ministry should consider making a request for a special budgetary allocation from the national government. The baseline analysis will be a source of solid evidence which can be used to justify the request. Other sources such as international donors should also be considered.

Stockpiling

Depending on the size and topography of the country, the Ministry of Health should consider prepositioning resources (facilities, staff, supplies and equipment) close to risk areas. This will speed up the delivery of tiered assistance, and avoid over-reliance upon a few centralized facilities and on transportation infrastructure that may themselves be damaged in the event of a natural or man-made major emergency.

The mobilization, transport and deployment of these stockpiled resources must be carefully planned.

Protection of the resources against natural and man-made hazards must be carefully considered in such plans, as well as the safety and security of staff and facilities.

Continuous monitoring of expiration of supplies and equipment is important in order to maintain quality and to permit updating and planning for future replenishment of these vital resources.

Evaluating first aider's response to the situation



Figure 11: First Aid: What is a First Aider?

http://2.bp.blogspot.com/sUtTekxc550/T9ySbkShbzI/AAAAAAAAAAM/nJTvZ_Uv_bg/s280/1.gif

First aid is the help given to someone who is injured or ill to keep them safe and to cause no further harm. The role of a first aider is to give someone this help. **Learn what to do.**

1. Assess the situation quickly and calmly.
 - **Safety:** check whether you or the casualty is in any danger. Is it safe to approach them? (Are you wearing protective equipment?)
 - **Scene:** find out what caused the accident or situation and how many casualties there are.
 - **Situation:** find out what's happened, how many people are involved and how old they are. Are there any children or elderly?
2. Protect yourself and them from any danger.
 - Always protect yourself first – never put yourself at risk.
 - Only move them if leaving them would cause them more harm.
 - If you can't make an area safe, call 999 or 112 for emergency help.
3. Prevent infection between you and them.
 1. Wash your hands with soap and water or rub your hands with alcohol gel.
 2. Wear disposable, latex free gloves. Do not touch an open wound without gloves on. If you do not have any gloves, you could use clean plastic bags or ask the casualty to dress the wound if they are able.
 3. Do not breathe, cough or sneeze over a wound or casualty.
4. Comfort and reassure.
 - Stay calm and take charge of the situation.
 - Introduce yourself to them to help gain their trust.

- Explain the situation and anything you're going to do before you do it.
 - Treat the casualty with dignity and respect at all times.
5. Assess the casualty and give first aid treatment.
- If there's more than one casualty make sure you help those with life-threatening conditions first.
 - Use the primary survey to deal with any life-threatening conditions. When these have been dealt with successfully move on to the secondary survey.
6. Arrange for help if needed.
- Call 999 or 112 for an ambulance if you think it's serious.

Qualities of a Good First Aider

There are a whole range of skills that are beneficial for a first aider to have in order to deal with situations in the best way possible. Some qualities of a good first aider are:

Communication Skills

If you are able to communicate effectively with the injured person, they will be much more likely to trust you and feel reassured. You will calm their sense of panic and make the situation a lot easier for you both. Having strong communication skills is also advantageous should you have to relay important information to the emergency services.

The Ability to Work under Pressure

The demands of a first aider can range anywhere from something simple, like a cut finger, to something a lot more serious, like a cardiac arrest. As a first aider you must be prepared for all possibilities, and know how to react quickly to the situation. You must keep calm – panicking will only make the situation worse – and be able to work effectively under pressure.

Initiative and Leadership

In an emergency, time can often play a crucial part in the outcome. Being able to act quickly, take the lead, use your initiative and trust your judgment are all vital skills.

The Ability to Work in a Team

As well as being able to lead, it is important that you can work effectively in a team. You may have to work alongside the emergency services, and being able to co-operate with them is vital.

Positivity

Someone who always sees the glass as half full rather than half empty is a prime candidate for a first aider. Positivity will help you to reassure the injured person and prevent yourself from quickly becoming mentally drained while on the job.

4.2.1.4 Learning Activities

4.2.1.4.1 Practical Activities

1. You work as a first aider in organization Y. You have received an emergency call from nearby highway that a public transport has been involved in an accident and the casualties are badly injured. Demonstrate your role as a first aider responding for that emergency once you reach at the scene of the accident?

4.2.1.4.2 Self-Assessment

You are provided with the following questions for self -assessment, attempt them and check your responses

1. You are a first aider working for a busy hospital in your county. The hospital management has asked you to prepare a planning process for evaluating response of the casualty to the intervention. Which important factor would you consider while designing the planning process?
2. The WHO's Health Action in Crises (HAC) has recently elaborated a five-year strategy to build the capacity of the health sector and communities in emergency preparedness and risk reduction. What are the four pillars of this strategy?
3. What are the guiding principles designed to assist planners, policymakers, and implementing agencies as they develop plans for emergency response and risk reduction?
4. No matter what forms that government and public administration take in addressing the emergency response and risk reduction, national emergency management systems should include which essential factors while planning for emergency response and risk reduction?
5. Why is it important to involve the Ministry of Health in all national emergency management potentially dealing with mass casualties?
6. What roles does the ministry of health play in the emergency management of disasters?

4.2.1.5 Tools, Equipment, Supplies and Materials

- Casualty department
- Functional skills laboratory
- First aid kit
- Emergency room
- Emergency room standard manuals
- Emergency trolley
- First Aid Manual.
- Tweezers
- Alcohol Swabs
- Antibiotic Ointment
- Bandages
- Gauze Pads

- Medical Tape
- Elastic Bandages
- Pain relievers
- Instant cold pack
- Waste disposal bins
- Personal protective equipment
- Health workers
- Presentation charts
- Flip charts

4.2.1.6 References

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4.2.1.7 MODEL ANSWERS

Self-Assessment

You are provided with the following questions for self -assessment, attempt them and check your responses

1. **You are a first aider working for a busy hospital in your county. The hospital management has asked you to prepare a planning process for evaluating response of the casualty to the intervention. Which important factor would you consider while designing the planning process?**
 - Project definition: determines the aim, objectives and scope of an emergency plan, and decides the tasks and the resources required to performing these tasks.
 - Formation of a representative planning group: to gather the required information and to gain the commitment of key people and organizations, both of which will contribute to the successful implementation of the plan.
 - Potential problem analysis: through breaking the problem into its components to examine risks, their causes, possible preventive strategies, response and recovery strategies, and trigger events for these strategies.
 - Resource analysis: to identify the required resources for the response and recovery strategies, resources available, discrepancy between requirement and availability, and responsibility.
 - Designation of roles and responsibilities to individuals and organizations.
 - Management structure concerning the command of individual organizations and control across organizations.
 - Developing strategies and systems for specific response and recovery.
 - Documentation: The written emergency plan will consist of the outputs of each step of the process.
2. **The WHO's Health Action in Crises (HAC) has recently elaborated a five-year strategy to build the capacity of the health sector and communities in emergency preparedness and risk reduction. What are the four pillars of this strategy?**
 - Assessing and monitoring baseline information on the status of health emergency preparedness and risk management at regional and country levels
 - Institutionalizing emergency preparedness and risk management in ministries of health and establishing an effective all-hazard/ whole-health programme for this purpose
 - Encouraging and supporting community-based emergency preparedness and risk management
 - Improving knowledge and skills in health emergency preparedness and response, and risk management.
3. **What are the guiding principles designed to assist planners, policymakers, and implementing agencies as they develop plans for emergency response and risk reduction?**
 - Clear lines of responsibility
 - All-hazard applicability

- Scalability
 - Whole-of-health
 - Knowledge-based
 - Multi-sectoral collaboration
 - National policies which enable local solutions
4. **No matter what forms that government and public administration take in addressing the emergency response and risk reduction, national emergency management systems should include which essential factors while planning for emergency response and risk reduction?**
- Identification of lines of authority, from the national to the local level
 - Financial arrangements for funding emergency work
 - Arrangements to ensure that government and community activities are maintained (for example, creation of parallel or “hardened” communications systems to take over if normal voice or data transfer systems are affected)
 - National stockpiling of appropriate resources (including provincial or state and local prepositioning of stockpiles)
 - Database of national experts for advice on specific problems
 - Protocols and formal arrangements for coordinated efforts with other countries, or between provincial/state governments within the country.
5. **Why is it important to involve the Ministry of Health in all national emergency management potentially dealing with mass casualties?**
- Because of its expertise in the setting of health-related standards and rules and procedures
 - Because of its normative role in the setting of health-related standards and rules and procedures
6. **What roles does the ministry of health play in the emergency management of disasters?**
- Inter-ministry cooperation
 - Establishing a solid legal basis
 - Establishing the baseline data for disaster management
 - Hazard analysis and risk management
 - Monitoring system capacity
 - Surveillance, epidemiology and data collection
 - Developing early warning systems
 - Establishing financial and material resources
 - Stockpiling- prepositioning resources (facilities, staff, supplies and equipment) close to risk areas.

3 4.3 Learning Outcome 4:

Wind-up first aid intervention

4.3.1 Introduction to the learning outcome

This learning outcome specifies the content of competencies required to wind-up first aid intervention. It includes documentation, waste management and report writing

4.3.2 Performance Standard

1. Documented first aid situation as per first aid principles
2. Cleared first aid area as per first aid principles
3. Disposed off materials in regard to SOPs
4. Stored recyclable materials as per manufacturer's instructions and workplace policy

4.3.3 Information Sheet

Definition of Terms

Casualty- the department of a hospital providing immediate treatment for emergency cases

First Aid- is the first and immediate assistance given to any person suffering from either a minor or serious illness or injury, with care provided to preserve life, prevent the condition from worsening, or to promote recovery

Emergency services- are the public organizations whose job is to take quick action to deal with emergencies when they occur, especially the fire brigade, the police, and the ambulance service.

Injury- also known as physical trauma is damage to the body caused by external force. This may be caused by accidents, falls, hits, weapons, and other causes.

Medical waste is defined as: potentially infectious **waste** materials generated at **health care** facilities, such as hospitals, clinics, physician's offices, dental practices, blood banks, and veterinary hospitals/clinics, as well as **medical** research facilities and laboratories.

Documentation

Introduction

Good record keeping is an important aspect of a health and social care professional's role, and it is a fundamental part of health care.

An accurate written record detailing all aspects of patient monitoring is important, not only because it forms an integral part of the of the provision of care or nursing management of the patient, but because it also contributes to the circulation of information amongst the different teams involved in the patient's treatment or care.

In a legal sense, documentation and record keeping is also there for the protection of the nurse or healthcare professional.

A well-kept record can protect the practitioner in instances where the legal defence of their actions is required. Documentation also ensures a matter of professionalization and proof of the improvement of practices.



Figure 12: Record Keeping and Documentation

https://cdn.ausmed.com/ausmed-blog-images/2018/05/20180528_body.jpg

Types of Record-Keeping Used in Healthcare

1. Hand-written records
2. Computer-based systems (electronic)
3. Some organizations or employers will use a combination of both.

You'll be expected to be able to comply with whatever requirements your employer or organization sets for record-keeping. That means you'll need to:

- Ensure that you are up to date on the information systems and tools in your workplace including their security, confidentiality and appropriate usage;
- Protect any passwords or details given to you to enable your access to any systems;
- Make sure written records are not left in public places where unauthorized people might see them (including any electronic systems or displays);
- Ensure that an entry is made in the patient's medical record whenever a health professional sees a patient.

Components of a patient's records include:

- Medical records
- Nursing records/progress notes
- Medication charts
- Laboratory orders and reports
- Vital signs observation charts
- Handover sheets and admission
- Discharge and transfer checklists/ letters
- Patient's assessment forms, such as nutrition or pressure area care assessment.

Principles of Good Record Keeping

Some key factors underpin good record keeping. The patient's records should:

- Be factual, consistent and accurate;
- Be updated as soon as possible after any recordable event;
- Provide current information on the care and condition of the patient;
- Be documented clearly in such a way that the text cannot be erased;
- Be consecutive and accurately dated, timed and all entries signed (including any alterations);
- All original entries should be legible. Draw a clear line through any changes and sign and date;
- Not include abbreviations, slang or jargon as not all workplaces or organizations will use the same terminology;
- Records must be stored securely and should only be destroyed following your local policy;
- Avoid meaningless phrases, speculation and offensive subjective statements/insulting or derogatory language;
- Identify the patient by recording patient's name, date of birth and hospital number on each page of the record (three approved identifiers) or follow your local policies on how to identify patient's records;
- Still be legible if photocopied or scanned.

Common Deficiencies in Record Keeping

Poor record keeping hampers the care that patients receive and makes it difficult for healthcare professionals to defend their practice.

The most common deficiencies in record keeping include:

- An absence of clarity
- Inaccuracies
- Spelling mistakes
- Missing information
- Failure to record action taken when a problem has been identified.

Benefits of Good Record Keeping

Record keeping is a tool for professional practice and one that should help the care process. It is not separate and not an optional extra to be fitted in if circumstances allow.

A record should be made as soon as possible after the patient is seen or the procedure is complete. It's important that accurate record is made in the patient's notes and should include interventions and any response to the interventions.

The Importance of Good Record Keeping are:

- Record keeping makes the continuity of care easier;
- Record keeping promotes better communication and dissemination of information between members of the multi-professional team;
- Helps to address complaints or legal processes;
- Supports clinical audit, research, allocation of resources and performance planning;
- Helps to identify risks and enables the early detection of complications;
- Supports patient care and patient-centered communication;
- Supports effective clinical judgement;
- Supports delivery of services;
- Helps improve accountability;
- Shows how decisions were made relating to the patient's care.

Legal Issues in Record Keeping

The patient's records are occasionally required as evidence before a court of law, or to investigate a complaint at a local, organization level.

Sometimes records may be requested by professional governing bodies when investigating claims related to misconduct

Therefore, you must take care about what you write. Not only will you be asked to formally explain your records in the event of, for instance, a complaint from a patient/client, but registered nurses have both a professional and legal duty of care.

It is therefore critical to keep up-to-date with the legal requirements and best practices of record-keeping, proving that:

- A comprehensive nursing assessment of the patient has been undertaken including care that has been provided and planned;
- Relevant information is included together with any actions that have been taken in response to changes in patients' conditions;
- The duty of care to the patient has been provided and that no acts or omissions have compromised a patient's safety;
- Arrangements have been made for the ongoing care of the patient.

Waste Management

Types of Waste



Figure 13: Medical Waste for Disposal - Infection Risk Editorial Photo - Image of rubbish, medical waste

<https://images.app.goo.gl/NkDZCPNspmoXbKYN7>

1. **Infectious waste:** waste contaminated with blood and other bodily fluids (e.g. from discarded diagnostic samples), cultures and stocks of infectious agents from laboratory work (e.g. waste from autopsies and infected animals from laboratories), or waste from patients with infections (e.g. swabs, bandages and disposable medical devices);
2. **Pathological waste:** human tissues, organs or fluids, body parts and contaminated animal carcasses;
3. **Sharps waste:** syringes, needles, disposable scalpels and blades, etc.;
4. **Chemical waste:** for example solvents and reagents used for laboratory preparations, disinfectants, sterilants and heavy metals contained in medical devices (e.g. mercury in broken thermometers) and batteries;
5. **Pharmaceutical waste:** expired, unused and contaminated drugs and vaccines;
6. **Cytotoxic waste:** waste containing substances with genotoxic properties (i.e. highly hazardous substances that are, mutagenic, teratogenic or carcinogenic), such as cytotoxic drugs used in cancer treatment and their metabolites;
7. **Radioactive waste:** such as products contaminated by radionuclides including radioactive diagnostic material or radio therapeutic materials; and

8. **Non-hazardous or general waste:** waste that does not pose any particular biological, chemical, radioactive or physical hazard.

The major sources of health-care waste are:

1. hospitals and other health facilities
2. laboratories and research centers
3. mortuary and autopsy centers
4. animal research and testing laboratories
5. blood banks and collection services
6. nursing homes for the elderly

Health Care Risks of Hospital Wastes

1. Potentially harmful microorganisms that can infect hospital patients, health workers and the general public
2. Drug-resistant microorganisms which spread from health facilities into the environment.
3. Sharps-inflicted injuries;
4. Toxic exposure to pharmaceutical products, in particular, antibiotics and cytotoxic drugs released into the surrounding environment, and to substances such as mercury or dioxins, during the handling or incineration of health care wastes;
5. Chemical burns arising in the context of disinfection, sterilization or waste treatment activities;
6. Air pollution arising as a result of the release of particulate matter during medical waste incineration;
7. Thermal injuries occurring in conjunction with open burning and the operation of medical waste incinerators; and
8. Radiation burns.

Environmental Impact of Waste Disposal and Treatment

1. The disposal of untreated health care wastes in landfills can lead to the contamination of drinking, surface, and ground waters if those landfills are not properly constructed.
2. The treatment of health care wastes with chemical disinfectants can result in the release of chemical substances into the environment if those substances are not handled, stored and disposed in an environmentally sound manner.
3. Incineration of waste has been widely practiced, but inadequate incineration or the incineration of unsuitable materials results in the release of pollutants into the air and in the generation of ash residue. Incinerated materials containing or treated with chlorine can generate dioxins and furans, which are human carcinogens and have been associated with a range of adverse health effects. Incineration of heavy metals or materials with high metal content (in particular lead, mercury and cadmium) can lead to the spread of toxic metals in the environment.
4. Only modern incinerators operating at 850-1100 °C and fitted with special gas-cleaning equipment are able to comply with the international emission standards for dioxins and furans.

5. Alternatives to incineration such as autoclaving, microwaving, steam treatment integrated with internal mixing, which minimize the formation and release of chemicals or hazardous emissions should be given consideration in settings where there are sufficient resources to operate and maintain such systems and dispose of the treated waste.

Waste management: Reasons for Failure

1. Lack of awareness about the health hazards related to health-care waste
2. Inadequate training in proper waste management
3. Absence of waste management and disposal systems
4. Insufficient financial and human resources and the low priority given to the topic are the most common problems connected with health-care waste.
5. Many countries either do not have appropriate regulations, or do not enforce them.

The Way Forward

The management of health-care waste requires increased attention and diligence to avoid adverse health outcomes associated with poor practice, including exposure to infectious agents and toxic substances.

Key elements in improving health-care waste management are:

1. Promoting practices that reduce the volume of wastes generated and ensure proper waste segregation;
2. Developing strategies and systems along with strong oversight and regulation to incrementally improve waste segregation, destruction and disposal practices with the ultimate aim of meeting national and international standards;
3. Where feasible, favoring the safe and environmentally sound treatment of hazardous health care wastes (e.g, by autoclaving, microwaving, steam treatment integrated with internal mixing, and chemical treatment) over medical waste incineration;
4. Building a comprehensive system, addressing responsibilities, resource allocation, handling and disposal. This is a long-term process, sustained by gradual improvements;
5. Raising awareness of the risks related to health-care waste, and of safe practices; and
6. Selecting safe and environmentally-friendly management options, to protect people from hazards when collecting, handling, storing, transporting, treating or disposing of waste.

WHO Response

- WHO developed the first global and comprehensive guidance document, *Safe management of wastes from health-care activities*, now in its second edition and more recently a short guide that summarizes the key elements.

Safe management of wastes from health-care activities

- The guide addresses aspects such as;
 1. Regulatory framework

2. Planning issues
 3. Waste minimization and recycling
 4. Handling of waste
 5. Storage and transportation
 6. Treatment and disposal options
 7. Training.
- The document is aimed at managers of hospitals and other health-care facilities, policy makers, public health professionals and managers involved in waste management.
 - In addition, as part of monitoring Sustainable Development Goal 6 on safely managed water and sanitation, the WHO/UNICEF Joint Monitoring Programme will regularly report on safe management of health care waste as part of wider monitoring efforts on water and sanitation in health care facilities.

Waste Segregation

- **Segregation means** separating different **wastes** into different color-coded **bins** with liners or sharps **containers** at locations where they are generated, and it is always the first and the most important activity in the hospital set up.

Color Coded Waste Bins

1. **Red bin**—you put highly infectious wastes like blood, blood products, tissues, specimens, blood giving sets, tooth, amputated body parts etc.
2. **Yellow bin**—you put infectious wastes like fluid giving sets, used gloves, empty syringes, vials, ampules, tubing, wipes, and empty IVs etc.
3. **White bin**—you put the food remains e.g. banana peels, fruit remains etc.
4. **Black Bin**—you put waste papers like drugs cartons and packages



Figure 14: Color Coded Waste Bins

<https://im.indiatimes.in/content/itimes/photo/2015/Jan/6/1420535398-color-coded-waste-bins-iabv20.jpg?w=1400&h=1050&cc=1>



Figure 15: Sharps Containers - Infection Prevention & Control Associates

<https://images.app.goo.gl/DuxmDTe5zws3MCBc7>

Treatment and Disposal of Medical Waste

The primary methods of treatment and disposal of medical waste are:

1. Incineration
2. Autoclaves
3. Mechanical/Chemical Disinfection
4. Microwave
5. Irradiation
6. Vitrification

1. Incineration

Incineration is the controlled burning of the medical waste in a dedicated incinerator. Among industry professionals, these units are often referred to as hospital/medical/infectious waste incinerators (HMIWIs).

2. Autoclaves for Medical Waste

Autoclaves are closed chambers that apply heat and sometimes pressure and steam, over a period of time to sterilize medical equipment. Autoclaves have been used for a century to

sterilize medical instruments for re-use. Surgical knives and clamps, for instance, are put in autoclaves for sterilization.

3. Mechanical Processing and Chemical Disinfection

Mechanical processing can be grinding, shredding, and/or compacting. Mechanical treatment does not kill pathogens or disinfect equipment, but it can reduce waste volume in preparation of further treatment or disposal.

Chemical disinfection, primarily through the use of chlorine compounds, kills microorganisms in medical waste and can often oxidize hazardous chemical constituents. Chlorine bleach has been used for disinfecting processes for years – you might use it to clean your underwear and kill the E.coli.

4. Microwave

Microwave radiation is used to treat wastewater sludge and as a heat source to treat medical waste. Microwave treatment units can be either on-site installations or mobile treatment vehicles. The processing usually includes front-end shredding of the waste, both to increase the efficacy of the microwave treatment and to reduce the volume of the end waste for disposal. If the waste is dry, water is introduced and the wet waste is introduced to the microwave chamber.

5. Irradiation

Irradiation disinfects waste by exposing it to gamma rays that are fatal to bacteria. A radioactive isotope of cobalt is employed. This is basically the same radiation source used for radiation treatment of cancer. In cancer treatment, radiation is intended to kill the malignant cells. In irradiation for sterilization of equipment or treatment of waste, the radiation is intended to kill pathogens.

6. Vitrification

Although it is rarely used, vitrification can be an effective treatment for medical waste. The solid waste is mixed in when glass is formed (vitrification means production of glass). The high temperatures kill pathogens and some combustible material may burn or pyrolyze, resulting in an off-gas. Remaining material is encapsulated in glass, which has a very low diffusivity. There is little danger of hazardous materials leaching out of glass in significant quantities. The vitrified waste can therefore be put in a landfill with confidence.

A report is a short, sharp, concise document which is written for a particular purpose and audience. It generally sets out and analyses a situation or problem, often making recommendations for future action. It is a factual paper, and needs to be clear and well-structured.

Reports generally involve presenting your investigation and analysis of information or an issue, recommending actions and making proposals.

Requirements for the precise form and content of a report will vary between organization and departments and in study between courses, from tutor to tutor, as well as between subjects, so it's worth finding out if there are any specific guidelines before you start.

There are many different types of reports, including business, scientific and research reports, but the basic steps for writing them are the same. These are outlined below.

Steps of Report Writing

- Step 1: Decide on the 'Terms of reference'
- Step 2: Decide on the procedure
- Step 3: Find the information
- Step 4: Decide on the structure
- Step 5: Draft the first part of your report
- Step 6: Analyze your findings and draw conclusions
- Step 7: Make recommendations
- Step 8: Draft the executive summary and table of contents
- Step 9: Compile a reference list
- Step 10: Revise your draft report

Step 1: Decide on the 'Terms of reference'

To decide on the terms of reference for your report, read your instructions and any other information you've been given about the report, and think about the purpose of the report:

- What is it about?
- What exactly is needed?
- Why is it needed?
- When do I need to do it?
- Who is it for, or who is it aimed at?

This will help you draft your Terms of reference.

Step 2: Decide on the procedure

This means planning your investigation or research, and how you'll write the report. Ask yourself:

- What information do I need?
- Do I need to do any background reading?
- What articles or documents do I need?
- Do I need to contact the library for assistance?
- Do I need to interview or observe people?
- Do I have to record data?
- How will I go about this?

Answering these questions will help you draft the procedure section of your report, which outlines the steps you've taken to carry out the investigation.

Step 3: Find the information

The next step is to find the information you need for your report. To do this you may need to read written material, observe people or activities, and/or talk to people.

Make sure the information you find is relevant and appropriate. Check the assessment requirements and guidelines and the marking schedule to make sure you're on the right track. If you're not sure how the marks will be assigned contact your lecturer.

What you find out will form the basis, or main body, of your report – the findings.

Step 4: Decide on the structure

Reports generally have a similar structure, but some details may differ. How they differ usually depends on:

- The type of report – if it is a research report, laboratory report, business report, investigative report, etc.
- How formal the report has to be.
- The length of the report.

Depending on the type of report, the structure can include:

- A title page.
- Executive summary.
- Contents.
- An introduction.
- Terms of reference.
- Procedure.
- Findings.
- Conclusions.
- Recommendations.
- References/Bibliography.
- Appendices.
- The sections, of a report usually have headings and subheadings, which are usually numbered

Step 5: Draft the first part of your report

Once you have your structure, write down the headings and start to fill these in with the information you have gathered so far. By now you should be able to draft the terms of reference, procedure and findings, and start to work out what will go in the report's appendix.

Findings- The findings are result of your reading, observations, interviews and investigation. They form the basis of your report. Depending on the type of report you are writing, you may also wish to include photos, tables or graphs to make your report more readable and/or easier to follow.

Appendices-- As you are writing your draft decide what information will go in the appendix. These are used for information that:

- Is too long to include in the body of the report, or
- Supplements or complements the information in the report. For example, brochures, spreadsheets or large tables.

Step 6: Analyze your findings and draw conclusions

The conclusion is where you analyze your findings and interpret what you have found. To do this, read through your findings and ask yourself:

- What have I found?
- What's significant or important about my findings?
- What do my findings suggest?

For example, your conclusion may describe how the information you collected explains why the situation occurred, what this means for the organization, and what will happen if the situation continues (or doesn't continue).

Don't include any new information in the conclusion.

Step 7: Make recommendations

Recommendations are what you think the solution to the problem is and/or what you think should happen next. To help you decide what to recommend:

- Reread your findings and conclusions.
- Think about what you want the person who asked for the report should to do or not do; what actions should they carry out?
- Check that your recommendations are practical and are based logically on your conclusions.
- Ensure you include enough detail for the reader to know what needs to be done and who should do it.

Your recommendations should be written as a numbered list, and ordered from most to least important.

Step 8: Draft the executive summary and table of contents

Some reports require an executive summary and/or list of contents. Even though these two sections come near the beginning of the report you won't be able to do them until you have finished it, and have your structure and recommendations finalized.

An executive summary is usually about 100 words long. It tells the readers what the report is about, and summarize the recommendations.

Step 9: Compile a reference list

This is a list of all the sources you've referred to in the report and uses APA referencing.

Step 10: Revise your draft report

It is always important to revise your work. Things you need to check include:

- If you have done what you were asked to do. Check the assignment question, the instructions/guidelines and the marking schedule to make sure.
- That the required sections are included, and are in the correct order.
- That your information is accurate, with no gaps.
- If your argument is logical. Does the information you present support your conclusions and recommendations?
- That all terms, symbols and abbreviations used have been explained.
- That any diagrams, tables, graphs and illustrations are numbered and labeled.
- That the formatting is correct, including your numbering, headings, are consistent throughout the report.
- That the report reads well, and your writing is as clear and effective as possible.

You might need to prepare several drafts before you are satisfied. If possible, get someone else to check your report.

4.3.4 Learning Activities

4.3.4.1 Practical Activities

1. You are in the casualty where you provide first aid services to patients on daily basis but there is a problem of waste segregation in your unit. Perform waste segregation in your unit according to WHO standard?

4.3.4.2 Self-Assessment

You are provided with the following questions for self -assessment, attempt them and check your responses

1. You are working in the emergency room in organization X. What are the principles of good record keeping?
2. What are the components of a patient's record?
3. What is the importance of good record keeping?
4. Hospital wastes if not well handled can pose a great danger to the health workers. What are some of the health care risks of hospital wastes?
5. What are some of the key elements in improving health-care waste management?
6. Medical wastes need prompt treatment and disposal using the most effective methods and technology to ensure a safe environment. As a health worker, what are the primary methods of treatment and disposal of medical wastes?
7. You have been tasked with the responsibility of writing a medical report about the operation in your unit. What are the steps of report writing that can ensure you write a good report?

4.3.5 Tools, Equipment, Supplies and Materials

- Casualty department
- Functional skills laboratory
- First aid kit
- Emergency room
- Emergency room standard manuals
- Emergency trolley
- First Aid Manual.
- Tweezers
- Alcohol Swabs
- Antibiotic Ointment
- Bandages
- Gauze Pads
- Medical Tape
- Elastic Bandages
- Pain relievers

- Instant cold pack
- Waste disposal bins
- Personal protective equipment
- Health workers
- Presentation charts
- Flip charts

4.3.6 References

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4.3.7 MODEL ANSWERS

Self-Assessment

You are provided with the following questions for self -assessment, attempt them and check your responses

- 1. You are working in the emergency room in organization X. What are the principles of good record keeping?**
 - Be factual, consistent and accurate;
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 - Provide current information on the care and condition of the patient;
 - Be documented clearly in such a way that the text cannot be erased;
 - Be consecutive and accurately dated, timed and all entries signed (including any alterations);
 - All original entries should be legible. Draw a clear line through any changes and sign and date;
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 - Records must be stored securely and should only be destroyed following your local policy;
 - Avoid meaningless phrases, speculation and offensive subjective statements/insulting or derogatory language;
 - Identify the patient by recording patient's name, date of birth and hospital number on each page of the record (three approved identifiers) or follow your local policies on how to identify patient's records;
 - Still be legible if photocopied or scanned.
- 2. What are the components of a patient's record?**
 - Medical records
 - Nursing records/progress notes
 - Medication charts
 - Laboratory orders and reports
 - Vital signs observation charts
 - Handover sheets and admission
 - Discharge and transfer checklists/ letters
 - Patient's assessment forms, such as nutrition or pressure area care assessment.
- 3. What is the importance of good record keeping?**
 - Record keeping makes the continuity of care easier;
 - Record keeping promotes better communication and dissemination of information between members of the multi-professional team;
 - Helps to address complaints or legal processes;
 - Supports clinical audit, research, allocation of resources and performance planning;
 - Helps to identify risks and enables the early detection of complications;
 - Supports patient care and patient-centered communication;

- Supports effective clinical judgement;
- Supports delivery of services;
- Helps improve accountability;
- Shows how decisions were made relating to the patient's care.

4. Hospital wastes if not well handled can pose a great danger to the health workers. What are some of the health care risks of hospital wastes?

- Potentially harmful microorganisms that can infect hospital patients, health workers and the general public
- Drug-resistant microorganisms which spread from health facilities into the environment.
- Sharps-inflicted injuries;
- Toxic exposure to pharmaceutical products, in particular, antibiotics and cytotoxic drugs released into the surrounding environment, and to substances such as mercury or dioxins, during the handling or incineration of health care wastes;
- Chemical burns arising in the context of disinfection, sterilization or waste treatment activities;
- Air pollution arising as a result of the release of particulate matter during medical waste incineration;
- Thermal injuries occurring in conjunction with open burning and the operation of medical waste incinerators; and
- Radiation burns.

5. What are some of the key elements in improving health-care waste management?

- Promoting practices that reduce the volume of wastes generated and ensure proper waste segregation;
- Developing strategies and systems along with strong oversight and regulation to incrementally improve waste segregation, destruction and disposal practices with the ultimate aim of meeting national and international standards;
- Where feasible, favoring the safe and environmentally sound treatment of hazardous health care wastes (e.g, by autoclaving, microwaving, steam treatment integrated with internal mixing, and chemical treatment) over medical waste incineration;
- Building a comprehensive system, addressing responsibilities, resource allocation, handling and disposal. This is a long-term process, sustained by gradual improvements;
- Raising awareness of the risks related to health-care waste, and of safe practices; and
- Selecting safe and environmentally-friendly management options, to protect people from hazards when collecting, handling, storing, transporting, treating or disposing of waste.

6. Medical wastes need prompt treatment and disposal using the most effective methods and technology to ensure a safe environment. As a health worker, what are the primary methods of treatment and disposal of medical wastes?

- Incineration
- Autoclaves
- Mechanical/Chemical Disinfection
- Microwave

- Irradiation
- Vitrification

7. You have been tasked with the responsibility of writing a medical report about the operation in your unit. What are the steps of report writing that can ensure you write a good report?

- Step 1: Decide on the 'Terms of reference'
- Step 2: Decide on the procedure
- Step 3: Find the information
- Step 4: Decide on the structure
- Step 5: Draft the first part of your report
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